

WELCOME TO SUGARCUBE™

Set-up and Control of the SC2-Mini/Phono



Thank You for purchasing the SugarCube SC2-Mini/Phono

Your experience of high-end audio is about to get a lot sweeter.

*Instructions for the recording functions are in a separate document: **SC2 Recording Guide v2.x***

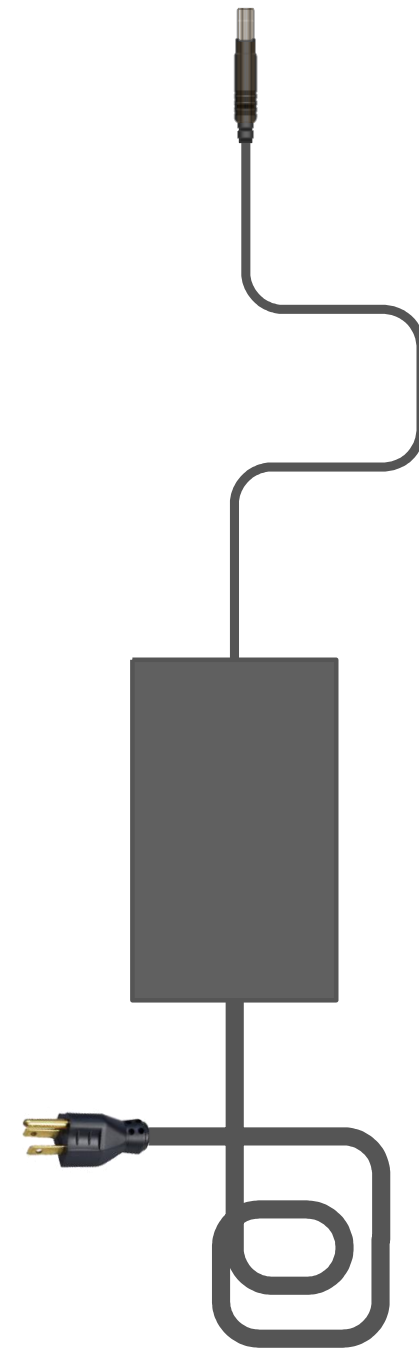
For any usage questions,
please contact:
support@sweetvinyl.com



1. Unpacking Your SugarCube



SugarCube SC2-Mini/Phono Unit



AC Power Adapter



Wifi USB Adapter



USB Stick

Please download the **SugarCube App** from your mobile device's app store to control your **SugarCube** remotely.

Adapter connects SugarCube to your WiFi network. Please use extension cable to route adaptor away from the back of the SugarCube. Alternatively, you can use wired Ethernet and not use the WiFi adaptor.

Stick contains the full SugarCube Setup Guide including mobile app and laptop control.

2. The SC2-Mini/Phono Front Panel Controls

RECORD

When pressed, SugarCube is recording audio. The **Ready/Rec** LED will turn from green to red.

STORAGE

USB thumb drive port accepts USB 3.0 media for transferring recordings. Instructions found in the SC2 Recording Guide.

PAIRING

Press knob to link your device with this SugarCube when asked for a pairing code (Pair). Instructions found in the manual.

NOTE: with software version 2.1 or greater, use code: **5123**

STATUS DISPLAY

Shows mode of operation, network status, software revision, etc.

CLICK REMOVE

When pressed and the LED is lit, SugarCube is processing audio to remove clicks and pops. The **BYPASS** LED will go out.

BYPASS

When pressed and the LED is lit, an internal relay has shunted the **LINE IN** directly to **LINE OUT**, completely bypassing all internal circuitry.

CLICK MON

When pressed and the LED is lit, the SugarCube suppresses the music so that you can better hear the clicks and pops. This helps set the **STRENGTH** level for particularly noisy LPs.

STRENGTH

Controls the strength of the click and pop detection and removal process. Almost all LPs will sound great at a setting mid-way.

Note: we recommend that you do NOT adjust **STRENGTH** all the way to maximum, you will rarely need this much. Sometimes less is more.



3. The SC2-Mini/Phono – Phono Setup

PHONO GROUND

Accepts the ground cable from your turntable.

PHONO INPUT

Connect your turntable to these ports.

MC CARTRIDGE LOADING

Continuous adjustment of MC cartridge loading from 20 Ohms to 5k Ohms.



MM CARTRIDGE GAIN

Gain boost: 0dB, +6dB, +12dB.
Additional MM cap. loading 300pF.



MM/MC SELECT

Switches between MM and MC cartridges. Out for MM & in for MC.

PHONO/LINE IN SELECT

Switches between LINE and PHONO. Out for PHONO & in for LINE.



4. The SC2-Mini/Phono – Connections



USB PORT FOR EXTERNAL DAC

Optional support for external USB DAC output.

USB PORT FOR WIFI ADAPTER

Or, plug the supplied Wifi adapter into a USB port.

POWER ON/OFF

Note: when power is off, an internal relay bypasses LINE IN to LINE OUT.

WIRED LAN

If an Ethernet LAN cable is available, connect to this port instead of plugging in the Wifi adapter.

LINE OUTPUT

Outputs to your line pre-amp. When the front panel BYPASS button is pressed, the audio from LINE IN passes directly to this LINE OUT through a mechanical audiophile-grade relay.

LINE INPUT

Accepts the output of your phono pre-amp. Please do NOT directly plug your turntable into this port unless it has a built-in phono pre-amp with a line-level output.

DC POWER IN

Connect the supplied adapter to this port.

Do either of the above when controlling the SC2-Mini/Phono on the SugarCube iOS or Android app (or desktop).

Connecting SugarCube to Your Sound System

1. Hardware Interconnects

**PLEASE BE SURE ALL
EQUIPMENT IS TURNED
OFF BEFORE ATTEMPTING
INSTALLATION**



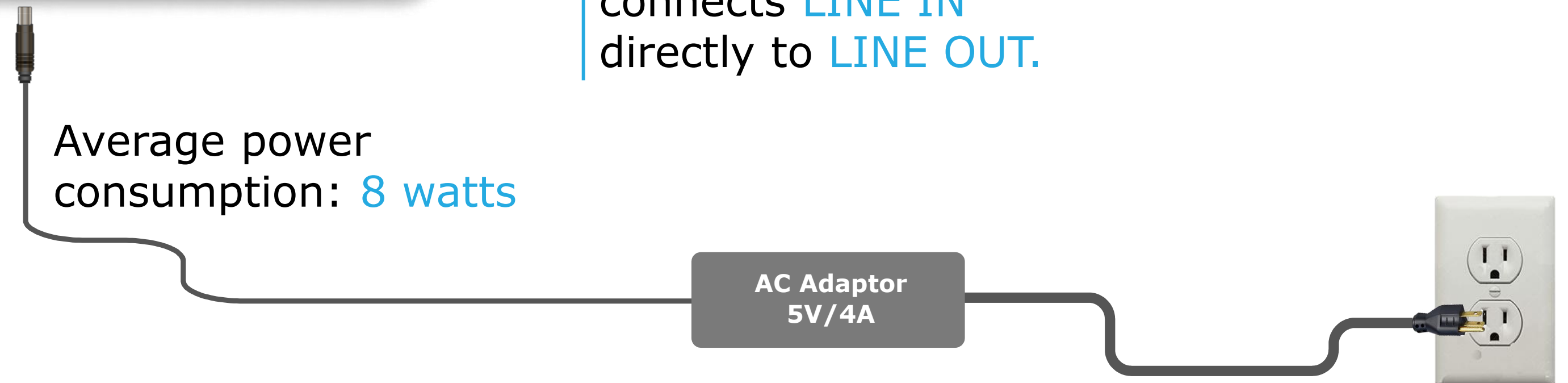
5. Connecting Your SC2-Mini/Phono



POWER ON/OFF

Note: when power is off, an internal relay connects **LINE IN** directly to **LINE OUT**.

Average power consumption: **8 watts**



Connecting the supplied AC adaptor.

1. Plug your SugarCube power adapter into a power source.
2. Please only use the supplied 5 volt adaptor or one that we approve.

6. The SC2-Mini/Phono – Phono Setup

Setting up Moving Magnet cartridges

MM CARTRIDGE GAIN
(Set L same as R)

Capacitive loading:
up=100pF; down=300pF

34dB

40dB

46dB

PHONO INPUT
Connect your turntable to these ports

Ground*

MC CARTRIDGE LOADING
This has no affect on MM cartridges

MM/MC SELECT
Switch should be OUT

PHONO/LINE IN SELECT
Switch should be OUT

*Note: some turntables such as Rega do not use a separate ground wire

SweetVinyl Mountain View, CA

SugarCube

Model: SC1-Mini/Phono

7. The SC2-Mini/Phono – Phono Setup

Setting up Moving Coil cartridges

MC CARTRIDGE GAIN

(Set R same as L)

Capacitive loading has no affect on MC

54dB

60dB

66dB

Connect your turntable to these ports

Ground*

MC CARTRIDGE LOADING

Continuous adjustment of MC cartridge loading from 20 Ohms to 5k Ohms.

Note: you can adjust this while the LP plays to listen for optimal setting, if you wish.



MM/MC SELECT

Press in

PHONO/LINE IN SELECT

Switch should be out

*Note: some turntables such as Rega do not use a separate ground wire



8. Connecting Your SC2-Mini/Phono

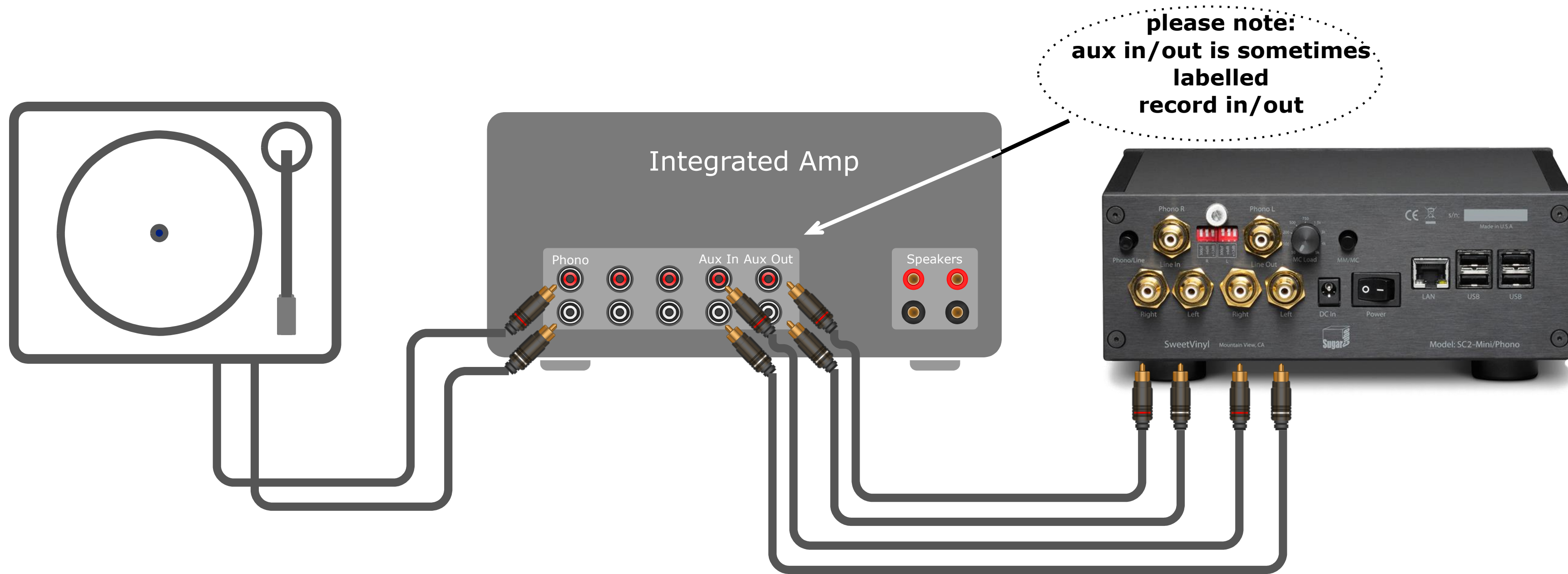


Many contemporary integrated amplifiers no longer have what was known as a 'tape loop' (back in the day of the cassette or reel-to-reel tape). Also, most modern home theatre processors that do have what are called 'effects loops' will not send audio from a phono input through the loop. In these cases, the internal phono stage on the SugarCube Mini/Phono allows direct connection to these devices.

The solution is to add a Mini/Phono stage (pre-amp) between your turntable and the Integrated Amp. Then connect the Line Out of the SugarCube to any available input on the integrated amp (except for the phono input, of course.)

9. Connecting Your SC2-Mini/Phono

Integrated Amp: using the 'tape loop'



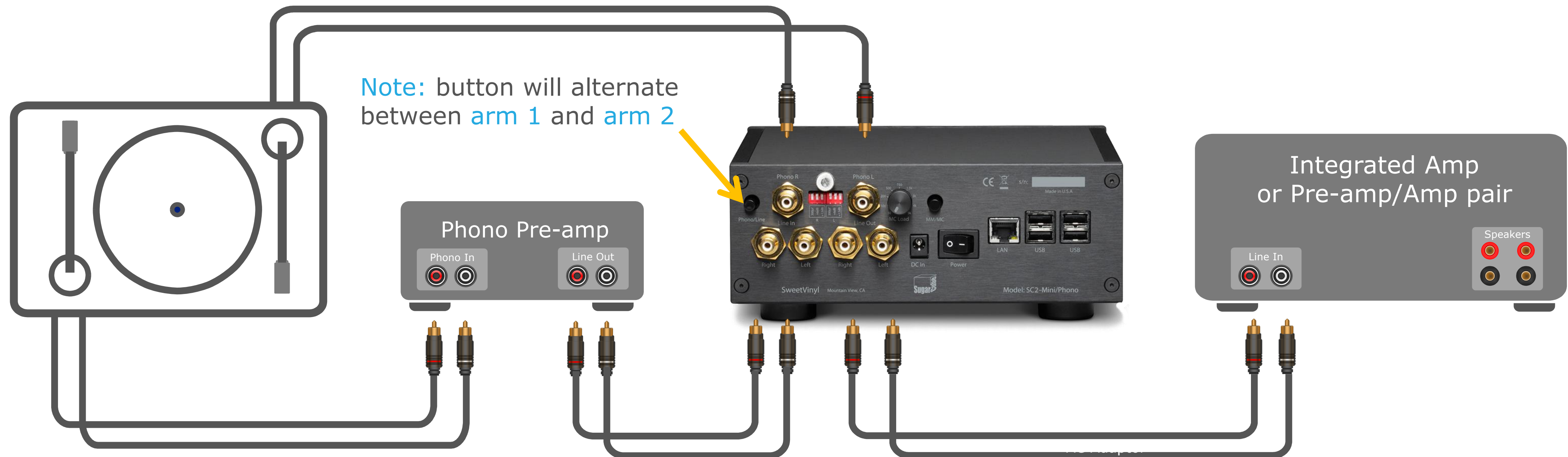
Connecting to an integrated amplifier via Aux In/Out (or Rec In/Out).

1. Plug your SugarCube power adapter into a power source.
2. Connect **AUX OUT** (or REC OUT) of your integrated amp (or processor) to the **LINE IN**.
3. Connect **LINE OUT** of the SugarCube to **AUX IN** (or **REC IN**) of your integrated amp or processor.
4. Select **LINE IN** using the back panel push button.

NOTE: DO NOT CONNECT THE **LINE OUT OF THE SC2-MINI/Phono TO THE **PHONO INPUT** ON YOUR EQUIPMENT.**

10. Connecting Your SC2-Mini/Phono

SETTING UP TWO TURNTABLES OR TWO ARMS



Connecting in series between a phono pre-amp and line-stage (pre-amp).

1. Plug your SugarCube power adapter into a power source.
2. Connect the output of your phono pre-amp to the **LINE IN** of the SugarCube.
3. Select **LINE IN** using the back panel push button.
4. Connect the **LINE OUT** of the SugarCube to an appropriate line-level input of your line-stage/pre-amp.

NOTE: DO NOT CONNECT THE **LINE OUT OF THE SUGARCUBE TO THE PHONO INPUT ON YOUR EQUIPMENT.**

11. Connecting Your SC2-Mini/Phono via WiFi



If you would like to use the SugarCube app to control your SugarCube, please connect either the WiFi adapter supplied with your SugarCube OR plug into a wired connection. There is no need to connect both except when setting up WiFi (please read appendix A).

- When using the WiFi adapter, once the SugarCube is powered on and operational, use the SugarCube app to set-up the WiFi connection by following the instructions on the app (iOS and Android operate similarly.)
- When connecting via wired Ethernet, you need only launch the SugarCube app and it will locate and connect to your SugarCube when the device running the app is on the same network with the SugarCube.

NOTE: THE SC2-Mini/Phono's basic functions of click and pop removal are fully available via front panel controls. However, the recording functions, other than start and stop recording, are only available via a table or laptop on a Google Chrome browser (instruction found in a separate document.)

12. Starting Up Your SugarCube



(1) SugarCube just powered on.



(2) SugarCube is ready to go.



(3) Hold the 'Bypass' button for a few seconds to see LAN address (and WiFi SSID if using wireless.)

Please note:

When you power up your SugarCube, it will take about two minutes to complete its startup procedures. During this time, the LCD display will show snow (our humorous tip-of-the-hat to the old days of analog TV). There is nothing wrong with your unit. We apologize for the wait but we have some seriously sophisticated software loading and it simply takes additional time. When you hear the results of our efforts, you will agree that it was worth the wait.

13. Pairing your SC2-Mini/Phono



Push in to PAIR

NOTE: with software version 2.1 or greater, you can use just use code: **5123**

In the course of setting up the connection between the SC2-Mini/Phono and the SugarCube app, or Web browser you will be prompted you to press and release the front panel knob to **PAIR**, then enter the four-digit code on the SugarCube display.

This button is also used when setting up WiFi for the SugarCube for use with the SugarCube app. When configuring WiFi, you will be instructed to press and HOLD the knob in for 5 SECONDS. The display of the SugarCube will announce that it is in WiFi setup mode, then continue setup on your mobile device, tablet, or desktop.

Setting Up SugarCube Mobile App

2. Connecting to the SugarCube

Apple iOS and Android
Guide





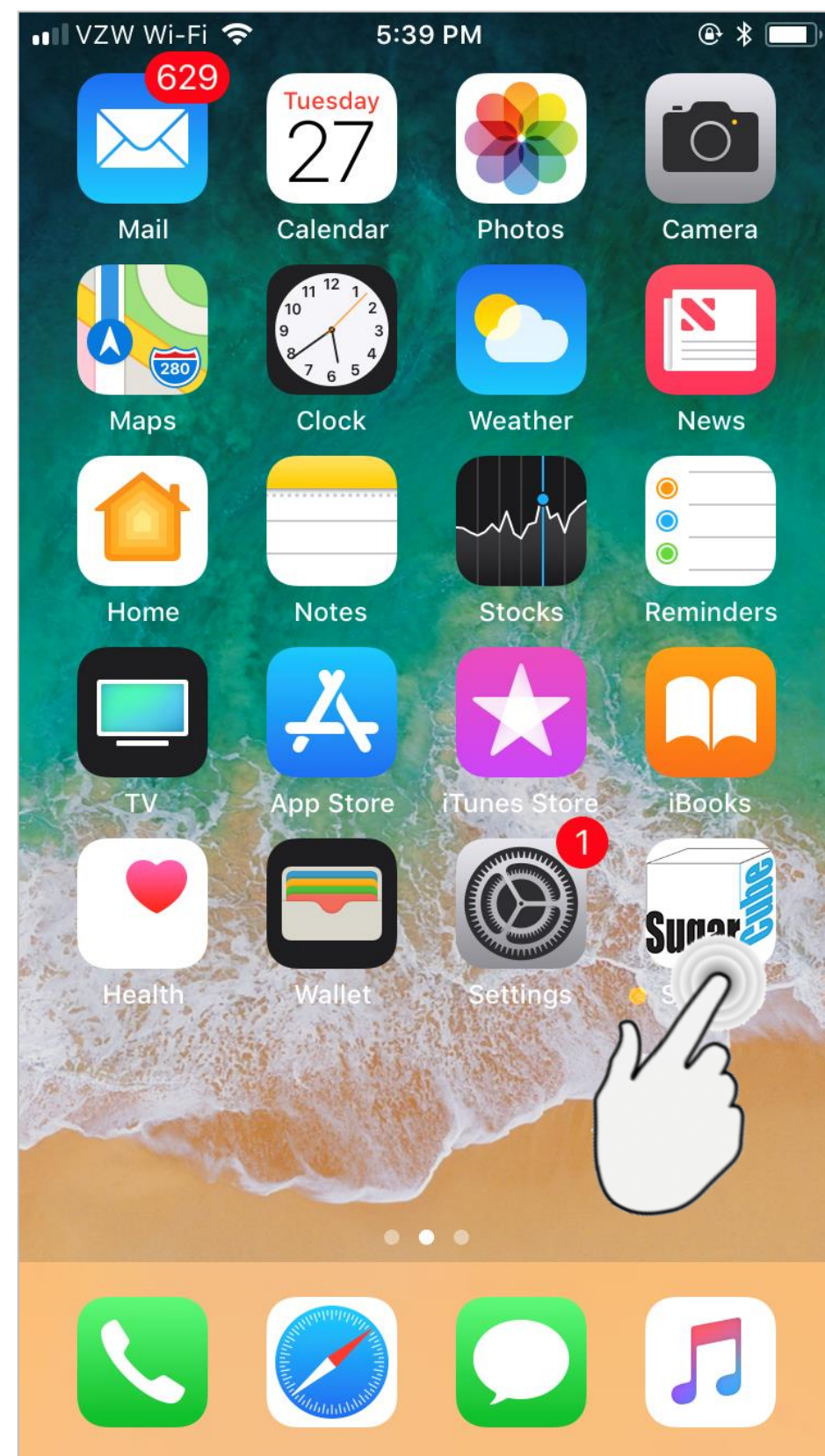
Using an Apple Mobile Device (Apple iOS)

14. Items needed to set up Apple to control SugarCube:

1. iPhone, iPad, Touch.
2. Network with WiFi your Apple device can communicate with and that can be connected to your SugarCube with WiFi or wired Ethernet.
3. From the Apple App Store, you will be able to download the 'SugarCube' app to your Apple mobile device. (Look for: SugarCube by SweetVinyl.)
4. The WiFi dongle supplied with your SugarCube.

Note: If you plan to use **wired Ethernet**, there is no need to plug in the WiFi adapter to the SugarCube; please skip to **Section 1.3** and follow the instructions there.

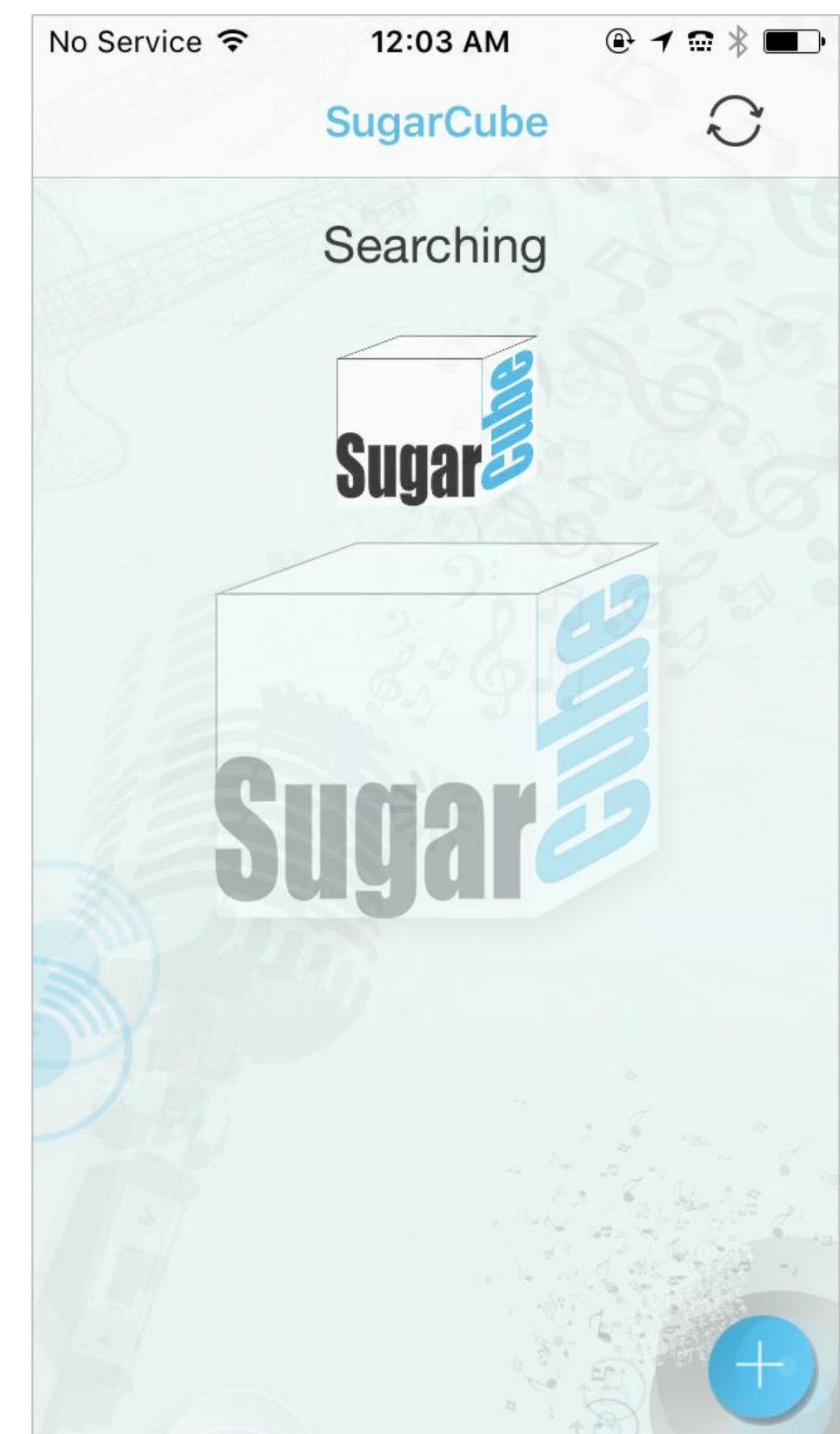
15. Connecting the app with SugarCube on Wi-Fi



(1) Launch the SugarCube app

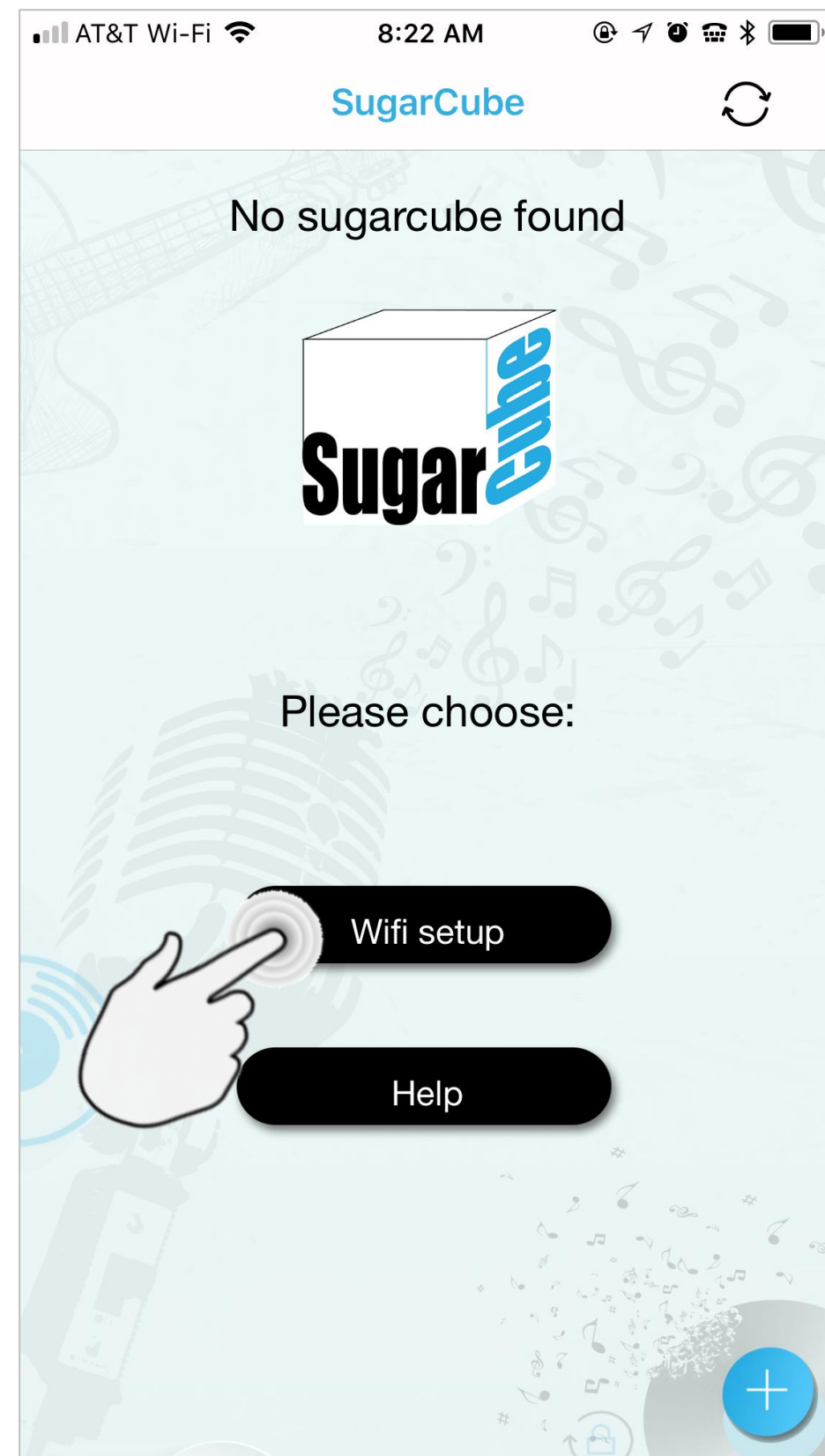


(2) App is loading

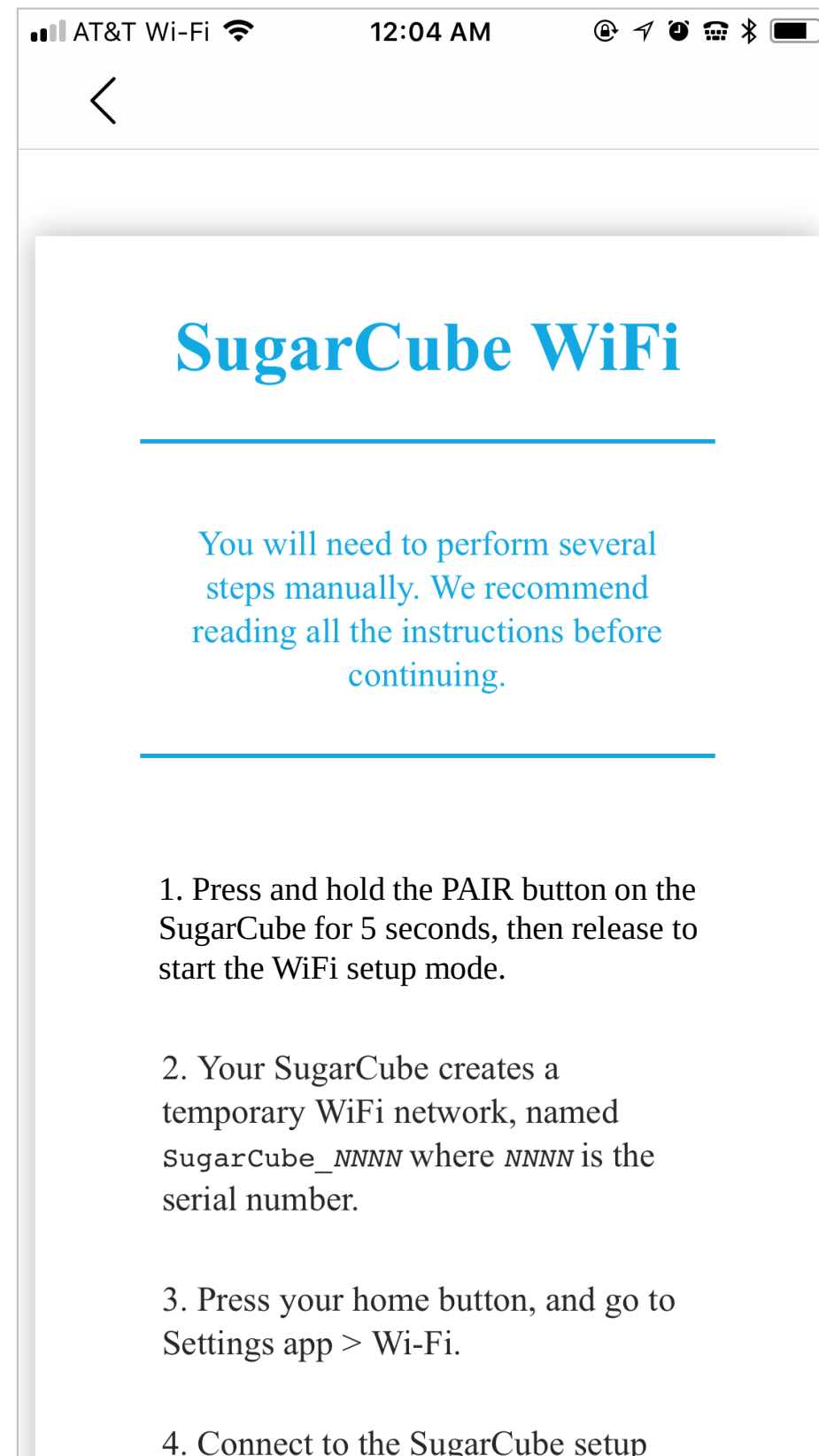


(3) App is searching for SugarCubes already setup on the LAN if any. For the first time user, nothing will be found, of course.

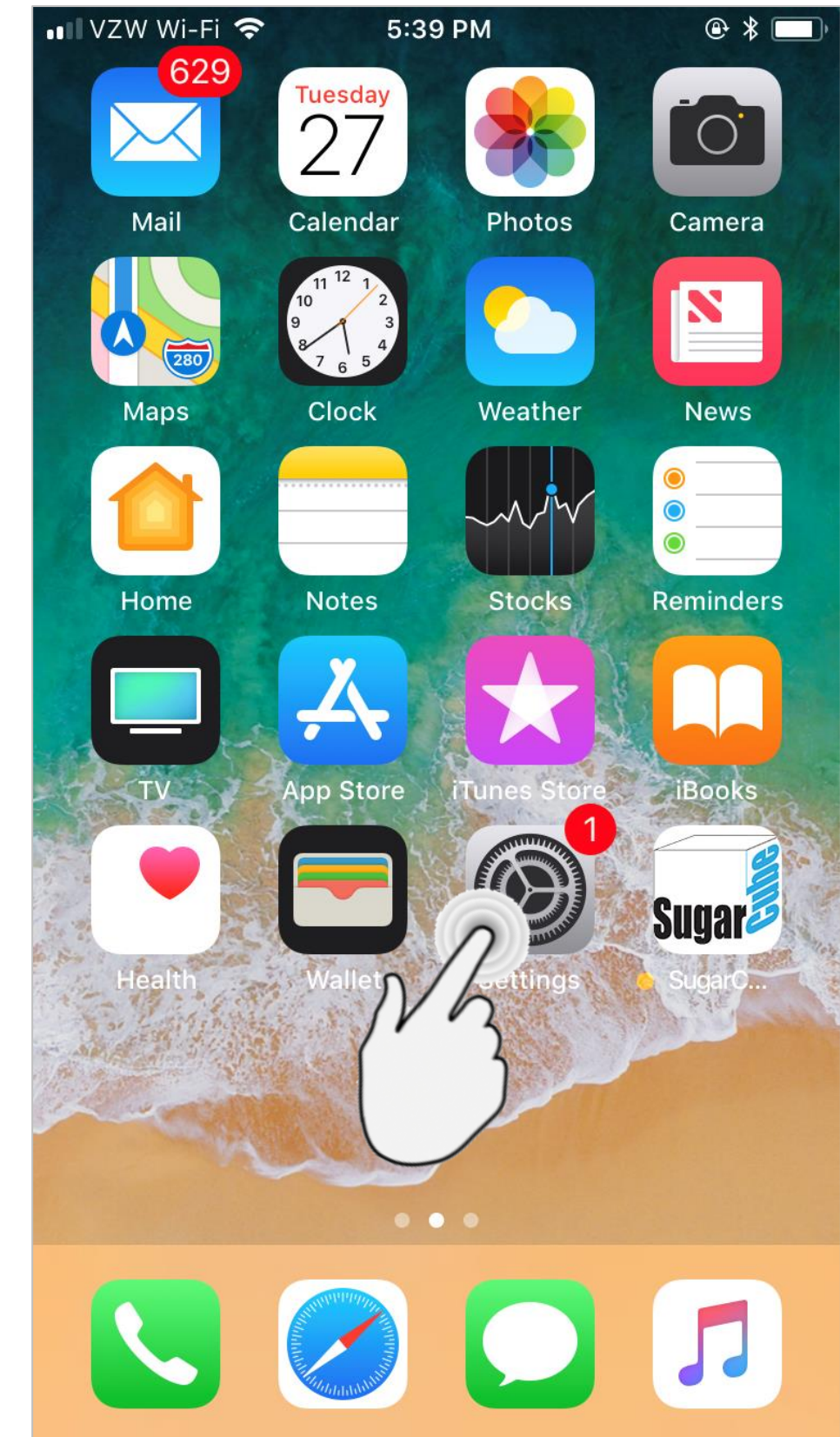
16. Connecting the app with SugarCube on Wi-Fi



(4) No SugarCube is found so the app offers the option to run WiFi setup.

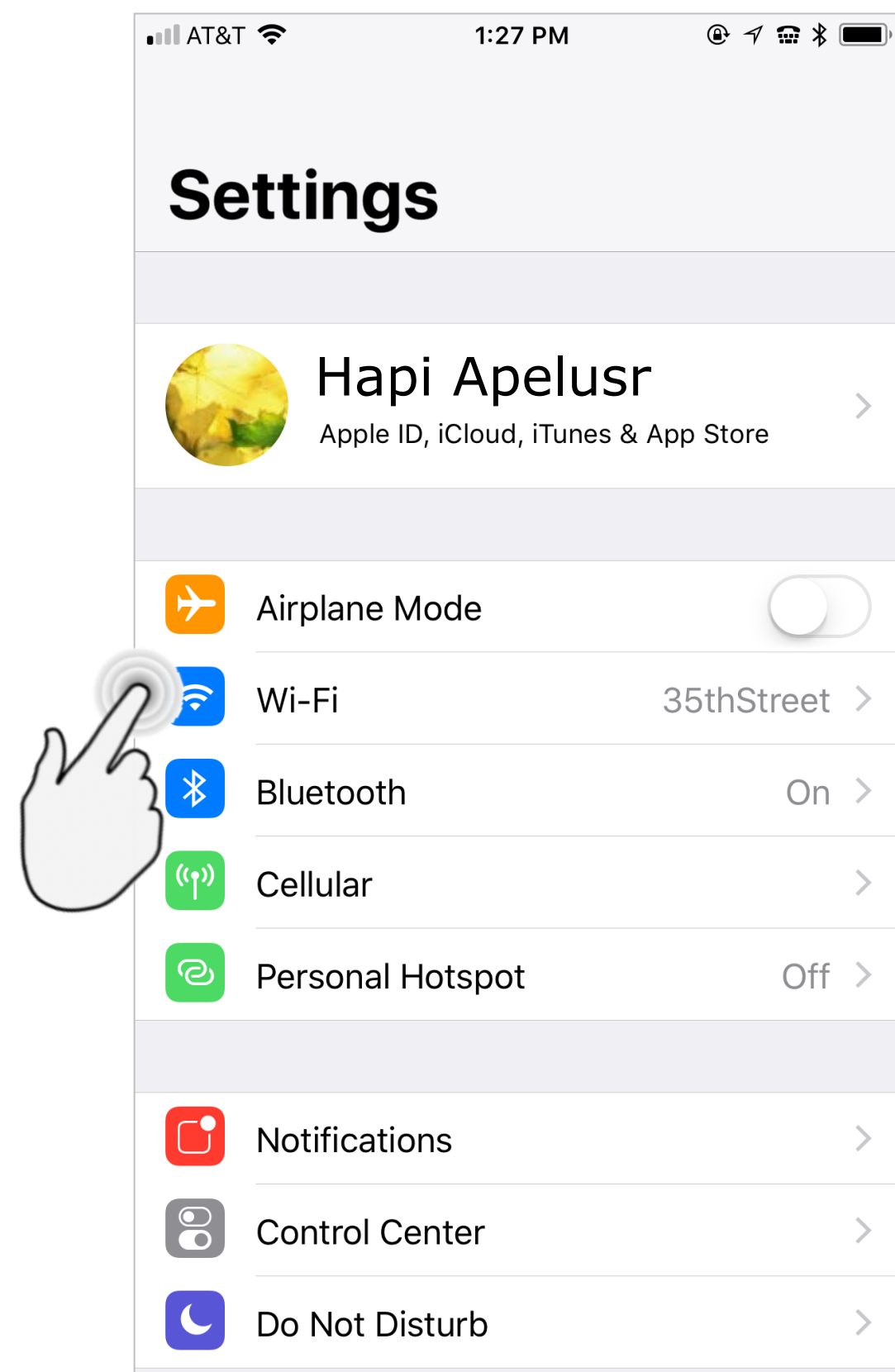


(5) The user is instructed to press and hold the front panel knob for 5 seconds to put the SugarCube into 'WiFi Setup Mode' which will be displayed on the front panel.

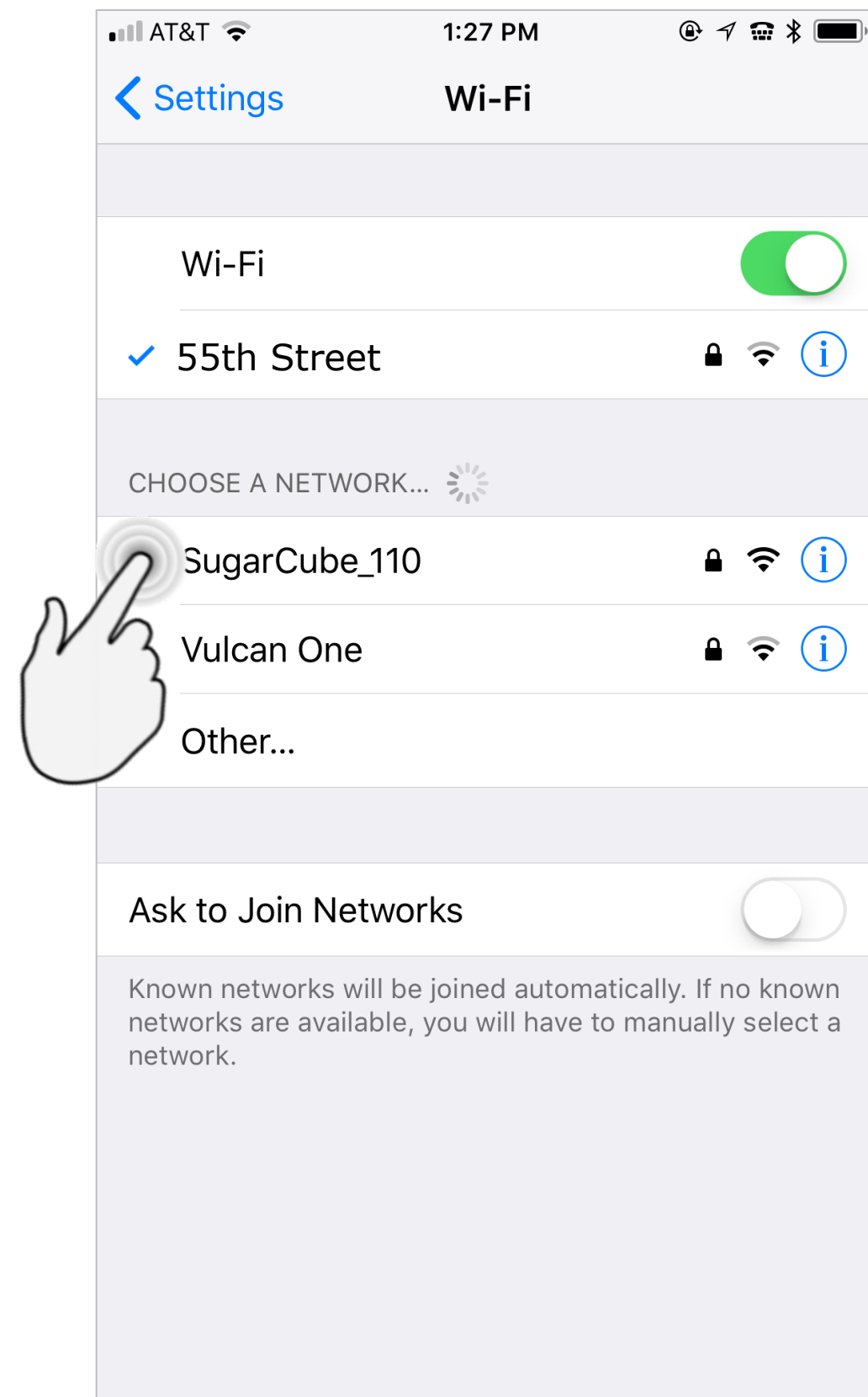


(6) The instructions direct the user to press the 'Home' key to get back to the 'Home' screen and then launch 'Setup'.

17. Connecting the app with SugarCube on WiFi

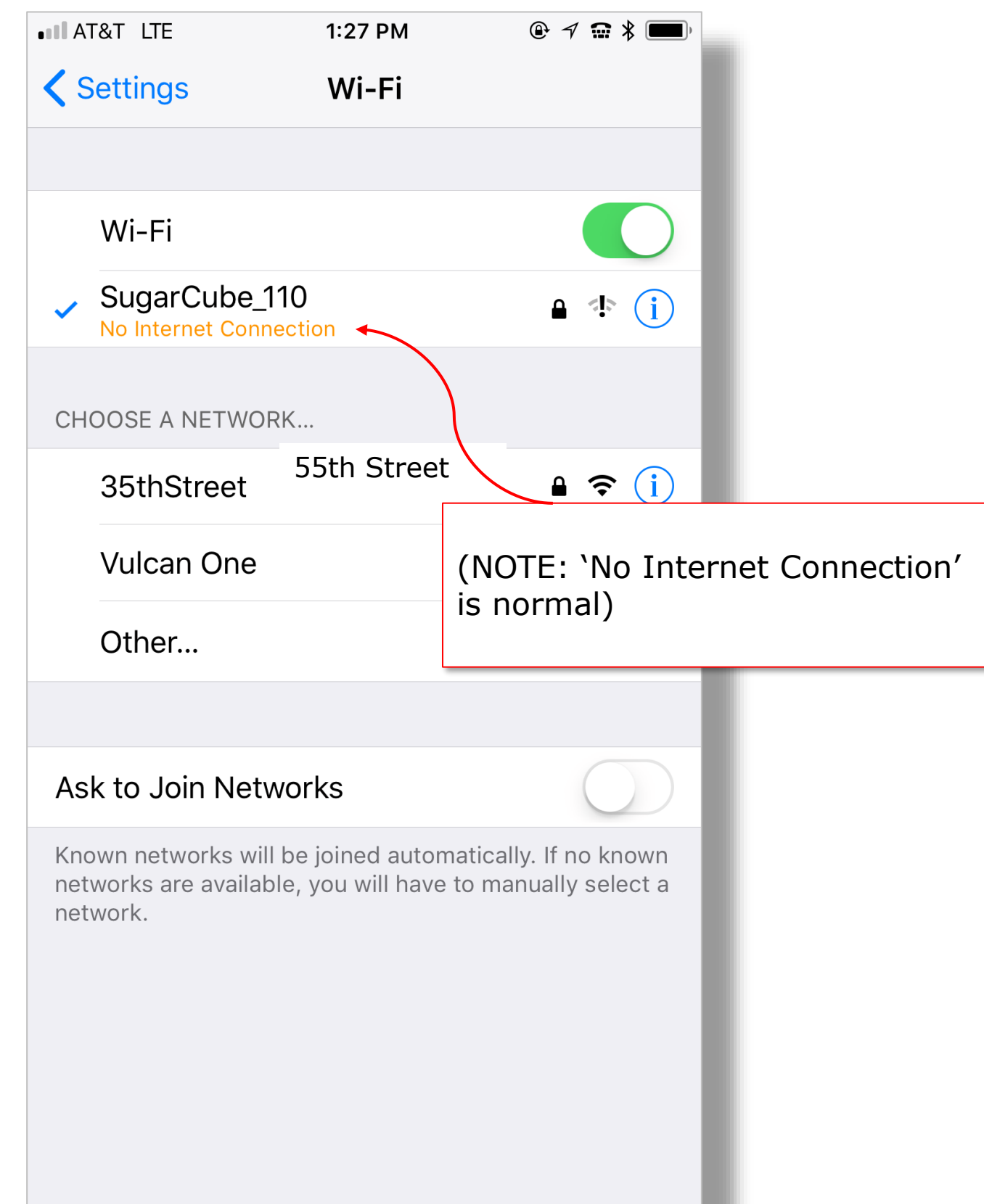


(7) When in 'Settings', select 'Wi-Fi'



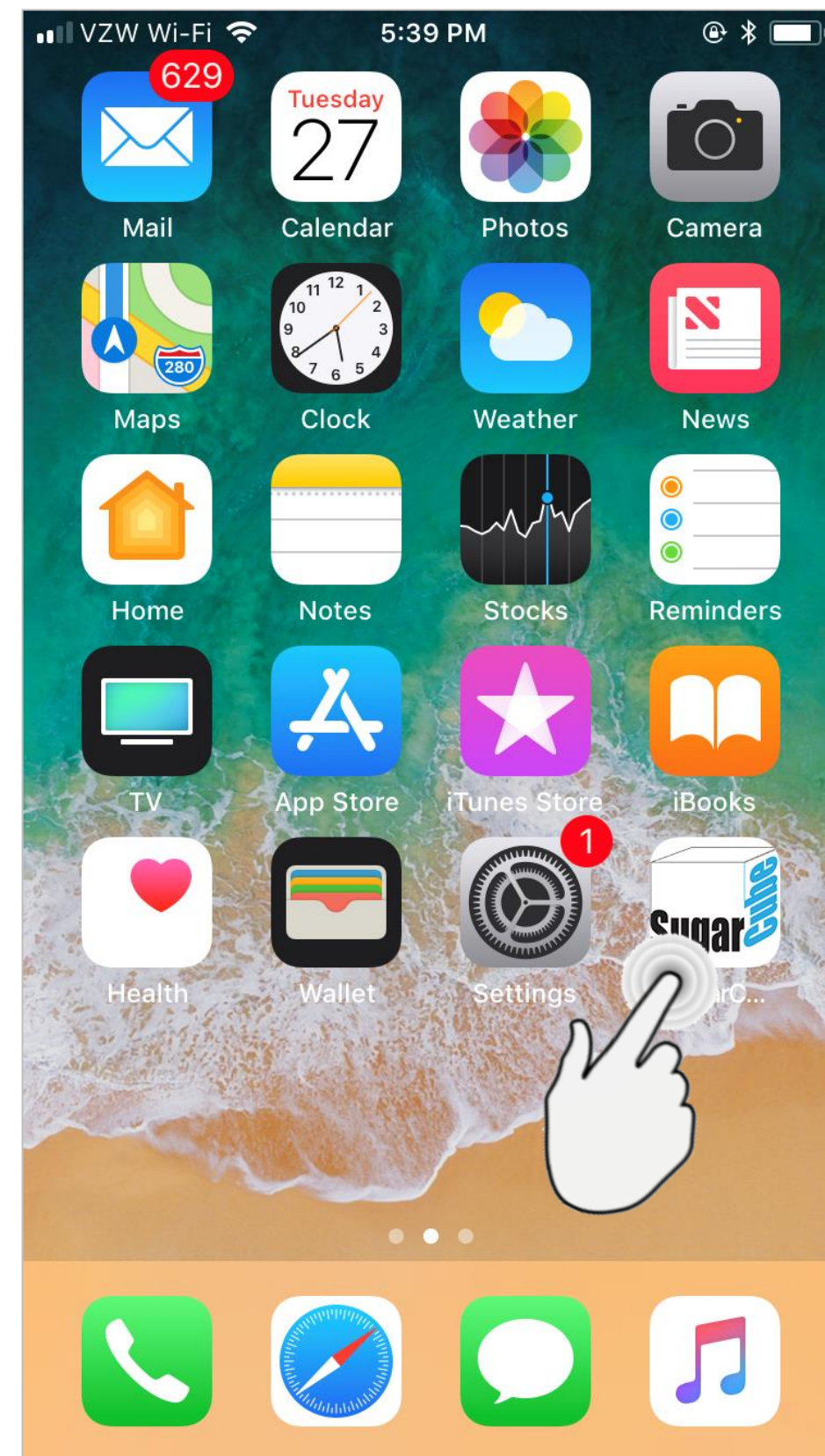
(8) Select 'SugarCube_nnnn', (whatever number appears after 'SugarCube_' is your device's serial number)

The password is: **SugarCube**

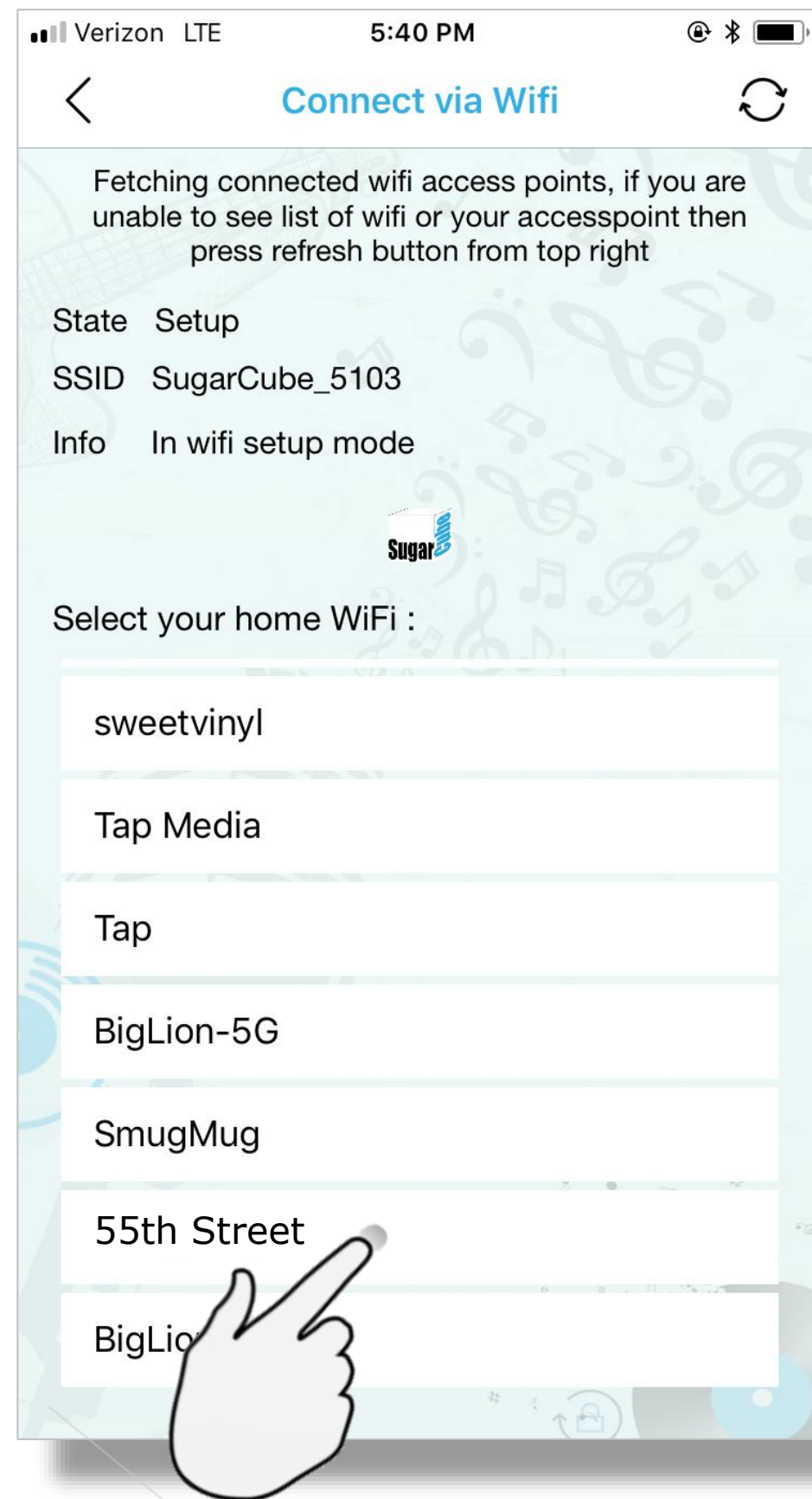


(9) With the 'SugarCube_nnnn' selected, your SugarCube is now temporarily acting as an access point.

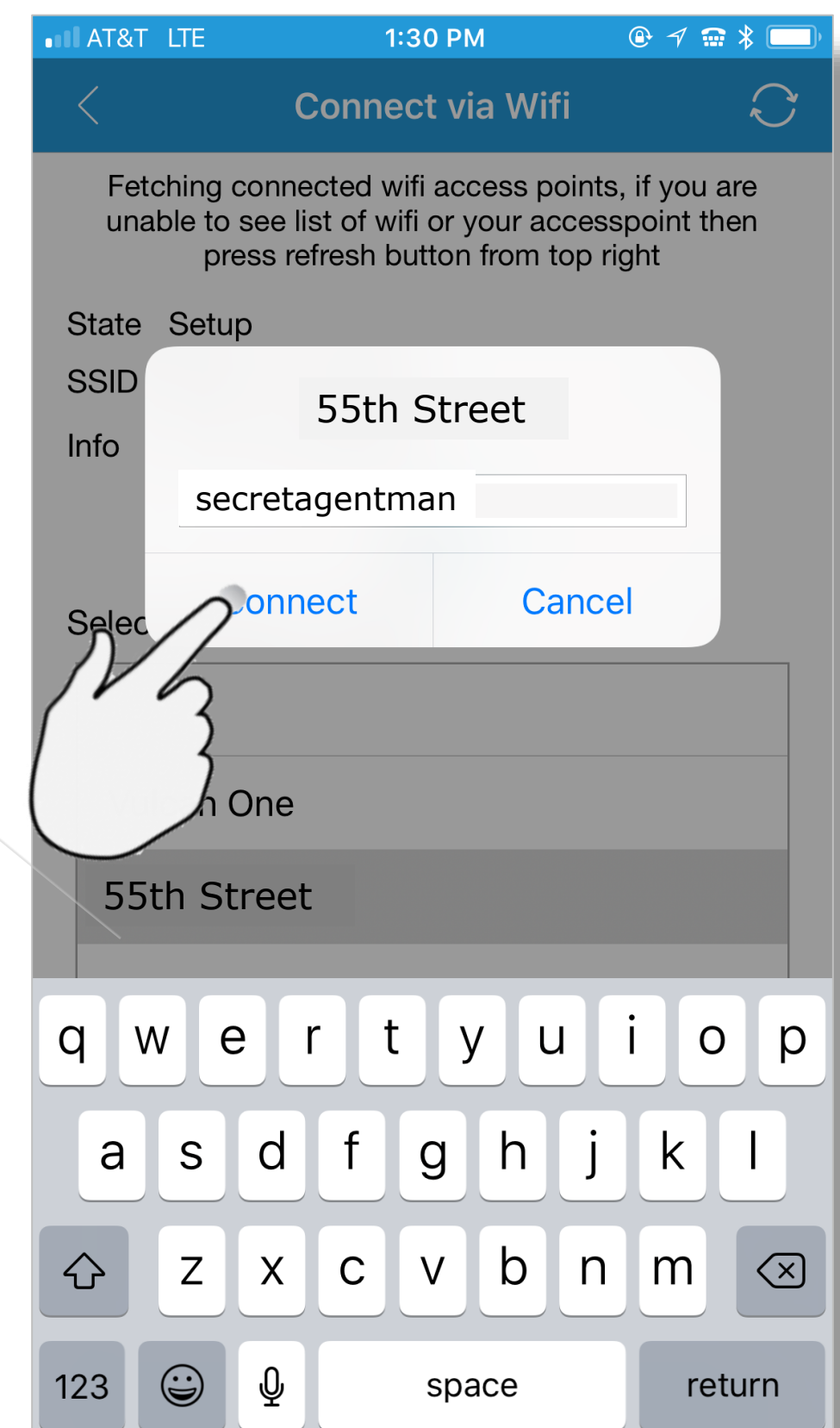
18. Connecting the app with SugarCube on Wi-Fi



(10) Relaunch the SugarCube app

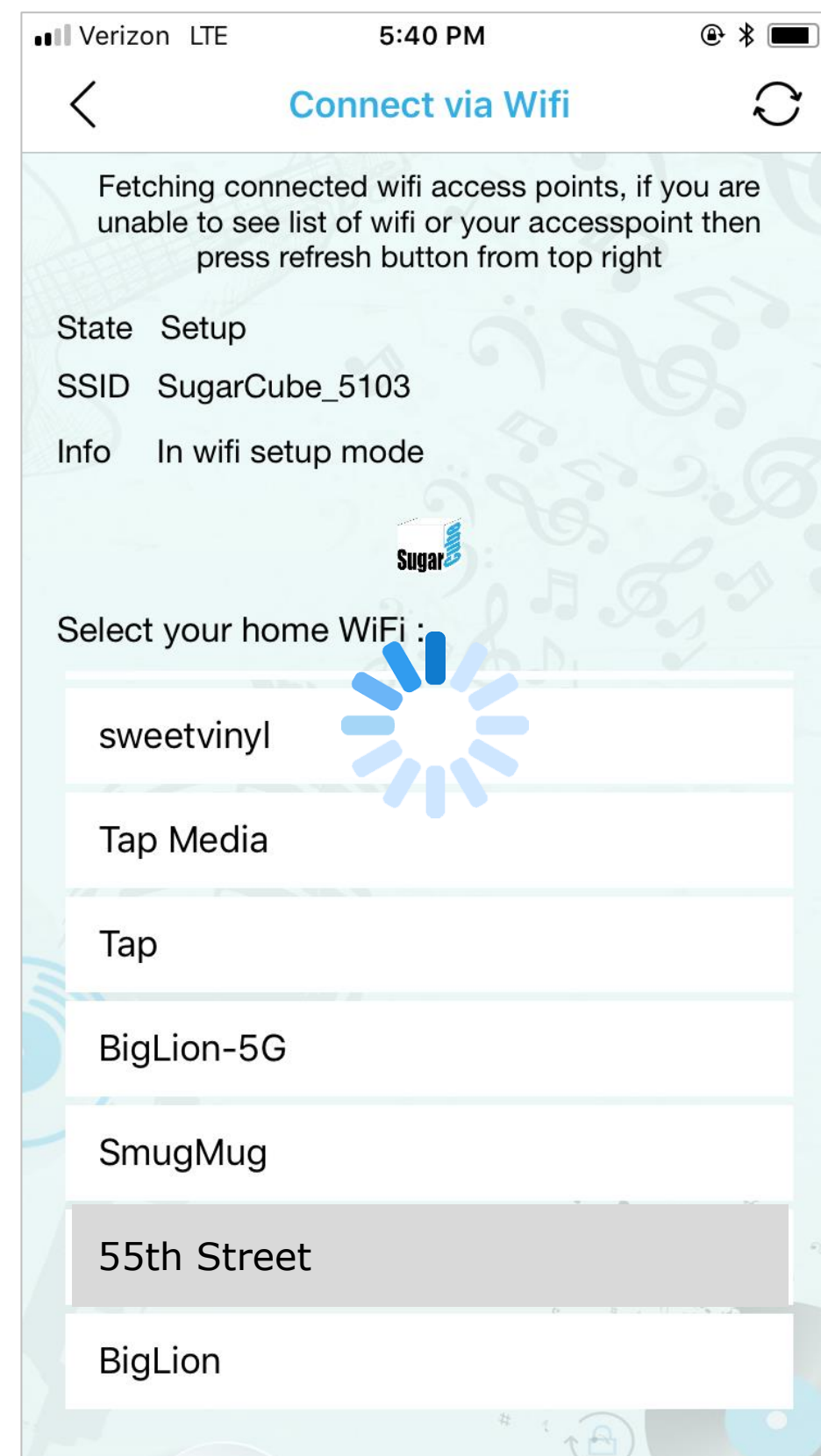


(11) Select the WiFi network that you wish to use



(12) Enter the WiFi password for your network

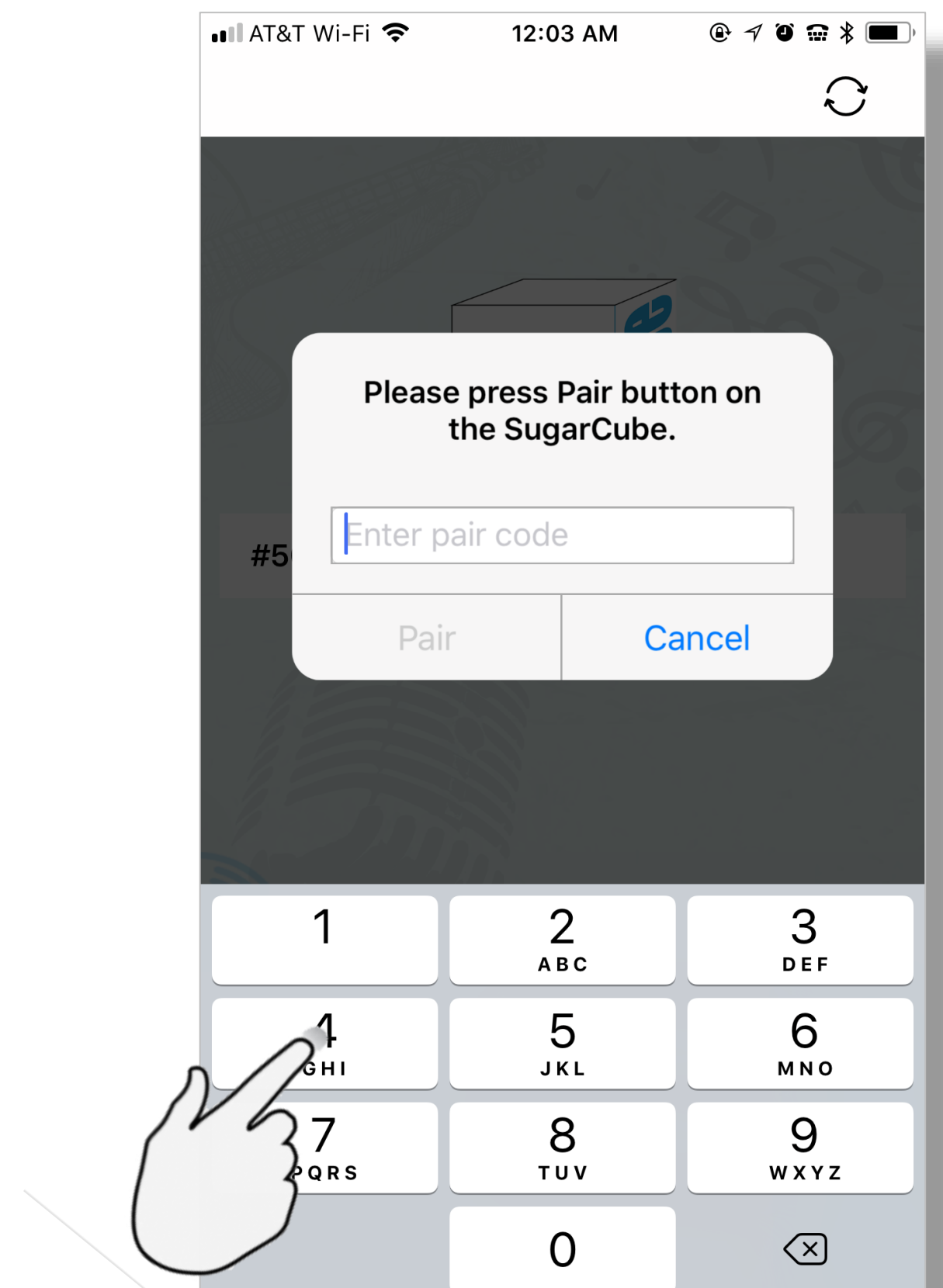
19. Connecting the app with SugarCube on Wi-Fi



(13) The app will now complete the connection to your WiFi network and also do a little housekeeping.

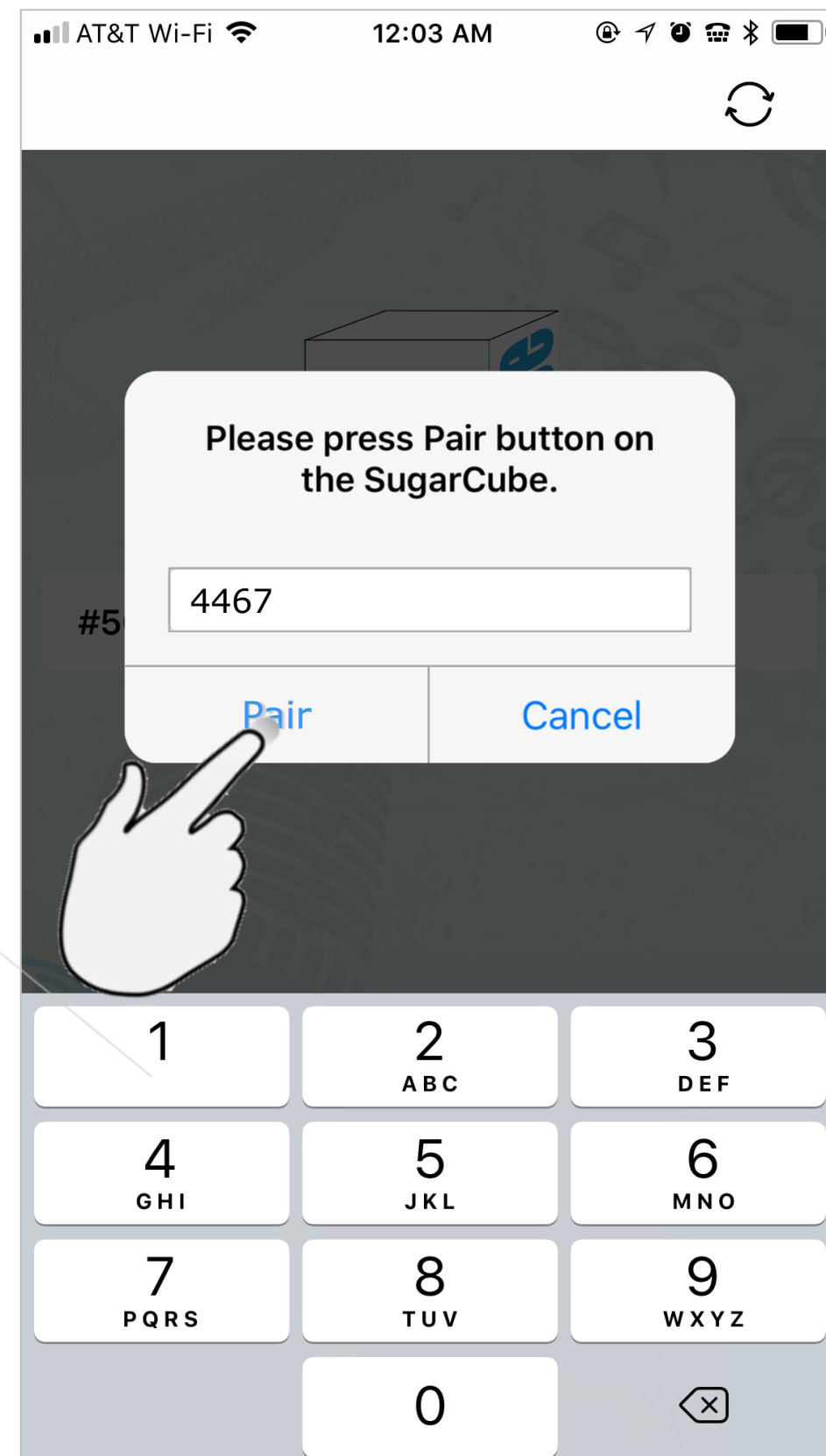


(14) The app will then display the SugarCube(s) found on your network. Select the appropriate device.

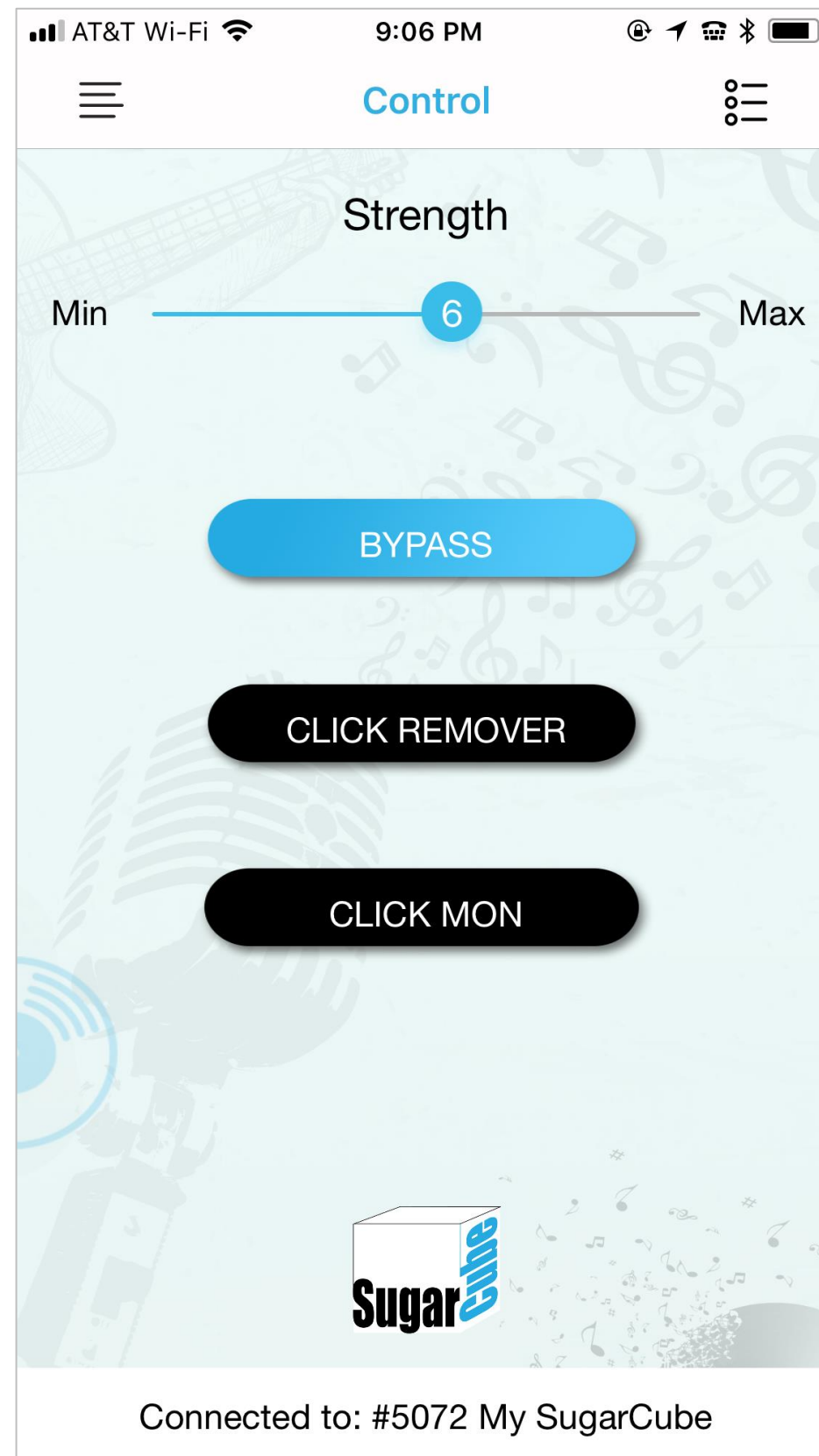


(15) If asked to 'Pair', press the 'front panel knob' of the SugarCube.

20. Connecting the app with SugarCube on Wi-Fi



(16) Enter the pairing code displayed on the SugarCube front panel and then press 'Pair' on the app screen.



(17) You will then see the 'Control' screen and you are done, congratulations!

NOTE: as of version 2.1 software, you can enter **5123** without needing to get a pair code.



Using an **Android** Mobile Device

21. Items needed to set up Android to control SugarCube:

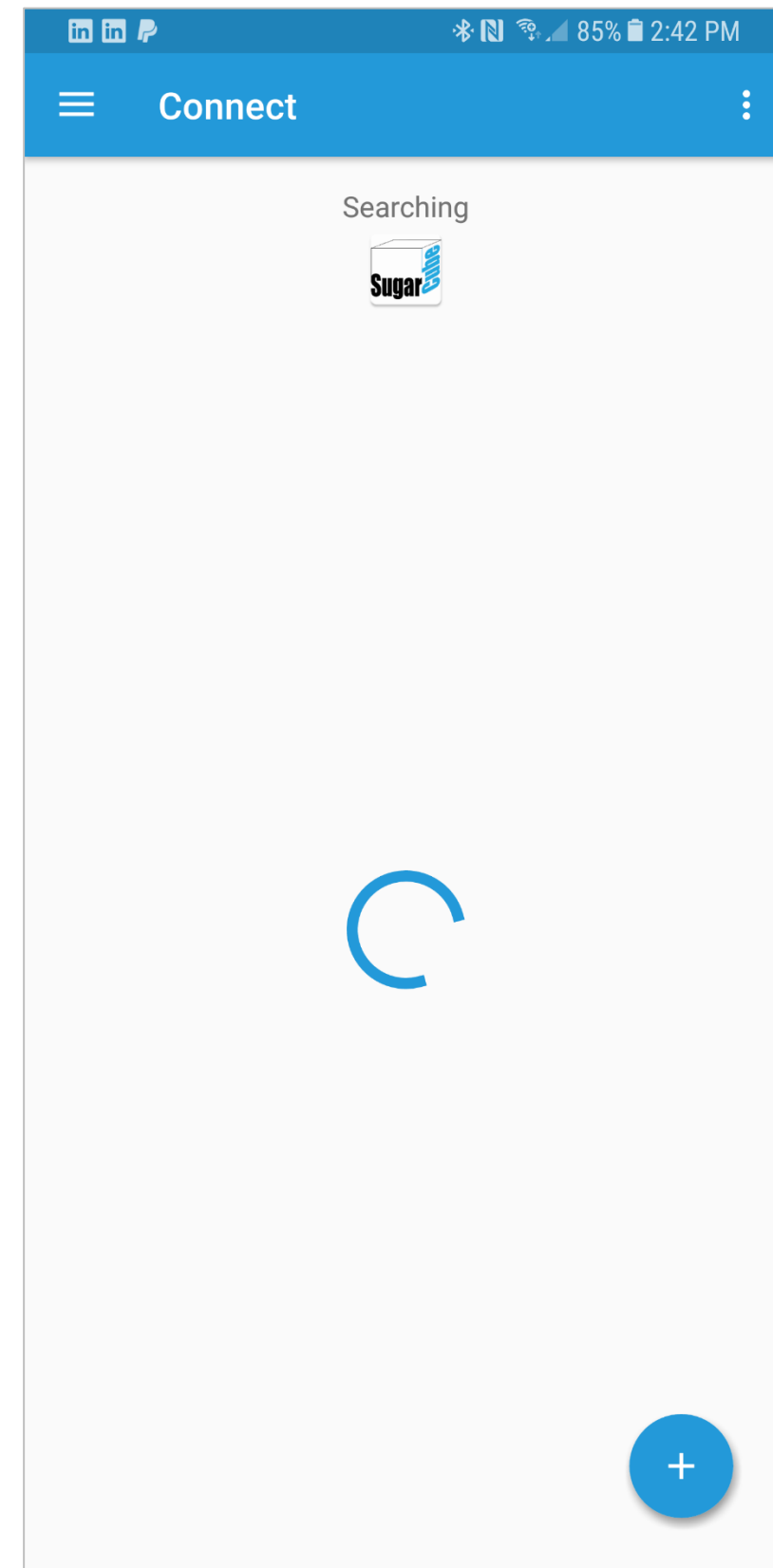
1. Android phone or tablet.
2. Network with WiFi your Android device can communicate with and that can be connected to your SugarCube with WiFi or wired Ethernet.
3. Download the Android App from the Google Play Store.
(Look for: SugarCube by SweetVinyl.)
4. The WiFi dongle supplied with your SugarCube or a wired Ethernet cable from a port on your router.

Note: If you plan to use wired Ethernet, there is no need to plug in the WiFi adapter to the SugarCube; please skip to page 1.10 and follow the instructions.

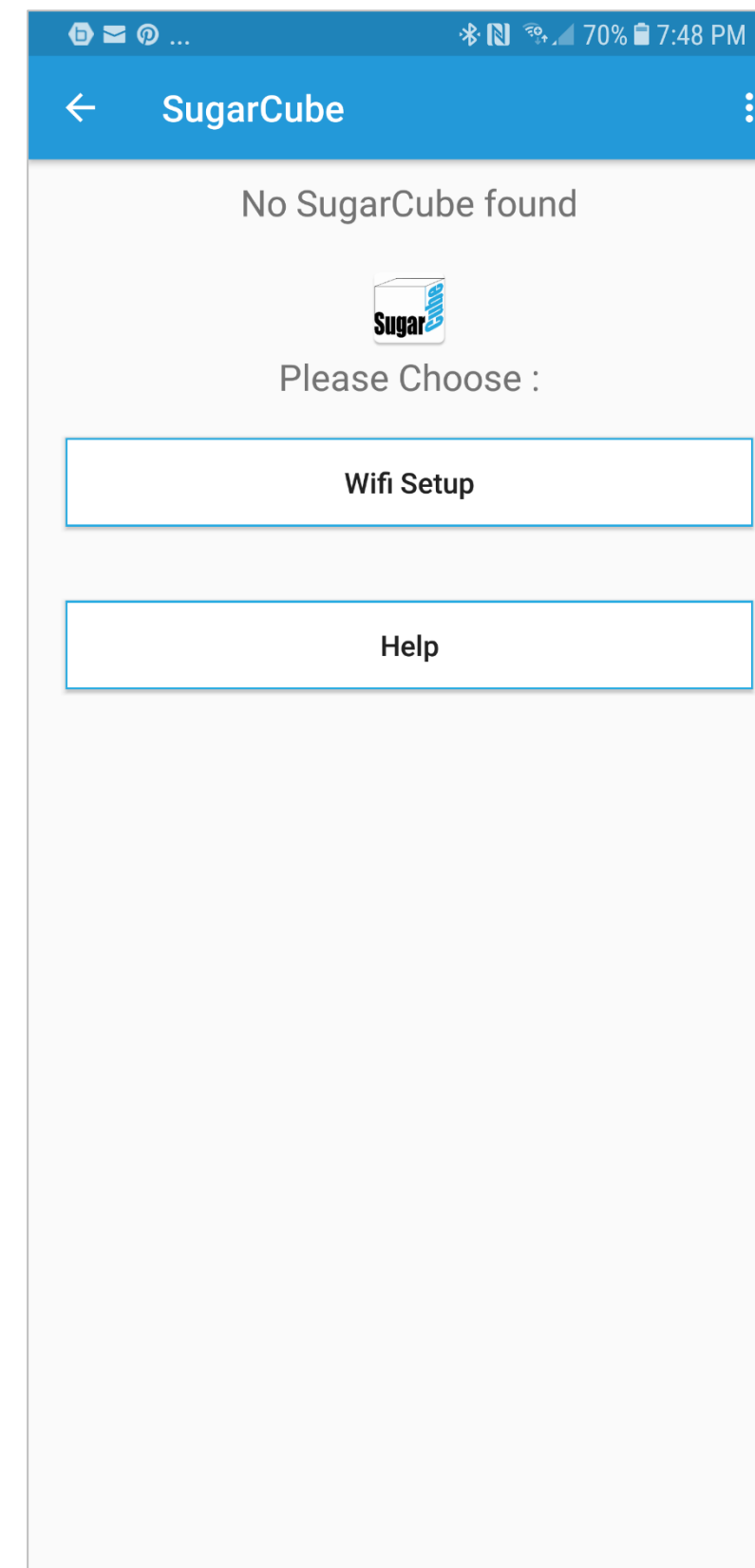
22. Connecting the app with SugarCube on Wi-Fi



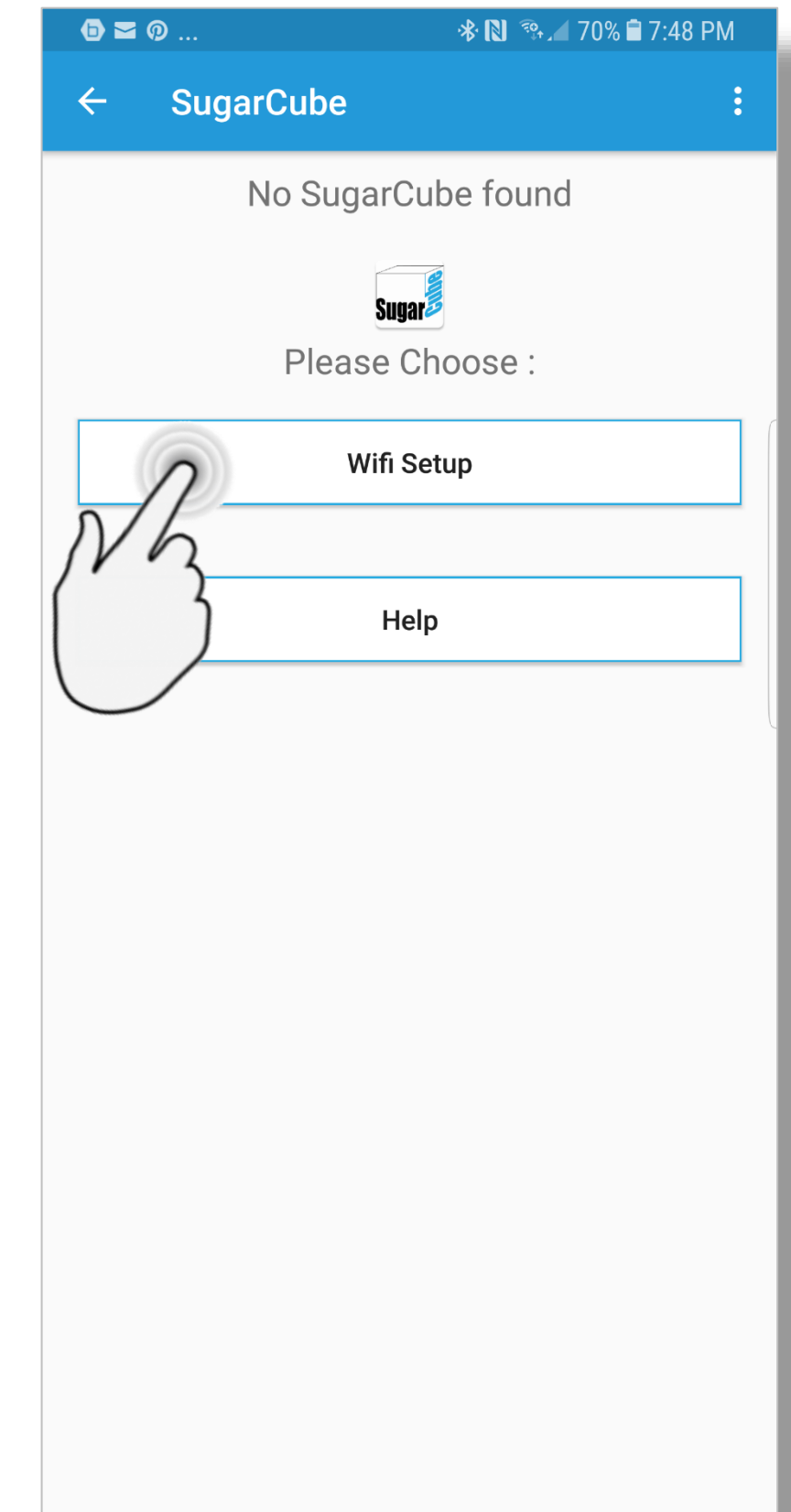
(1) App is loading



(2) App is searching for SugarCubes on the LAN

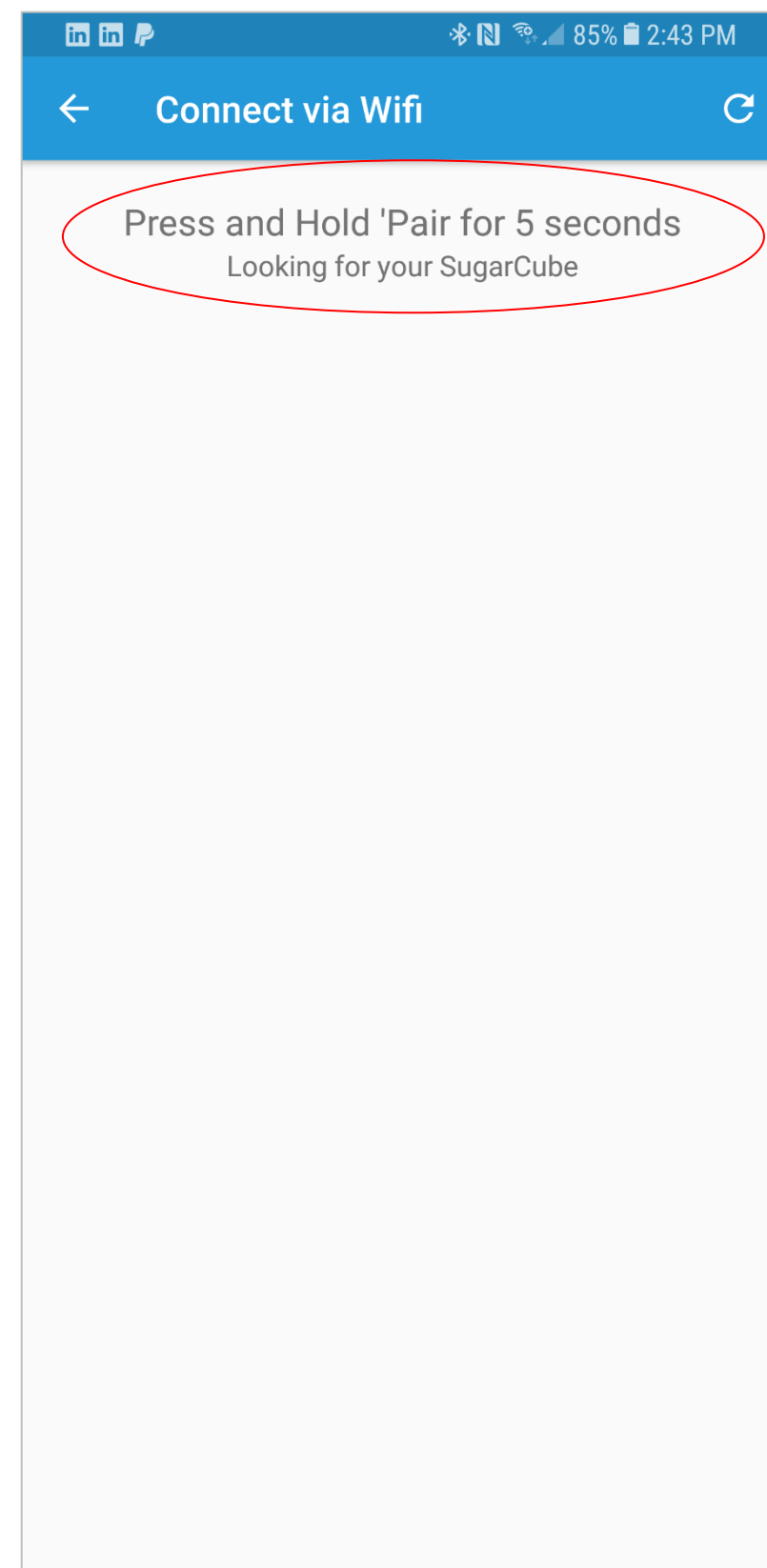


(3) No SugarCube is found so the app offers the option to run WiFi setup.

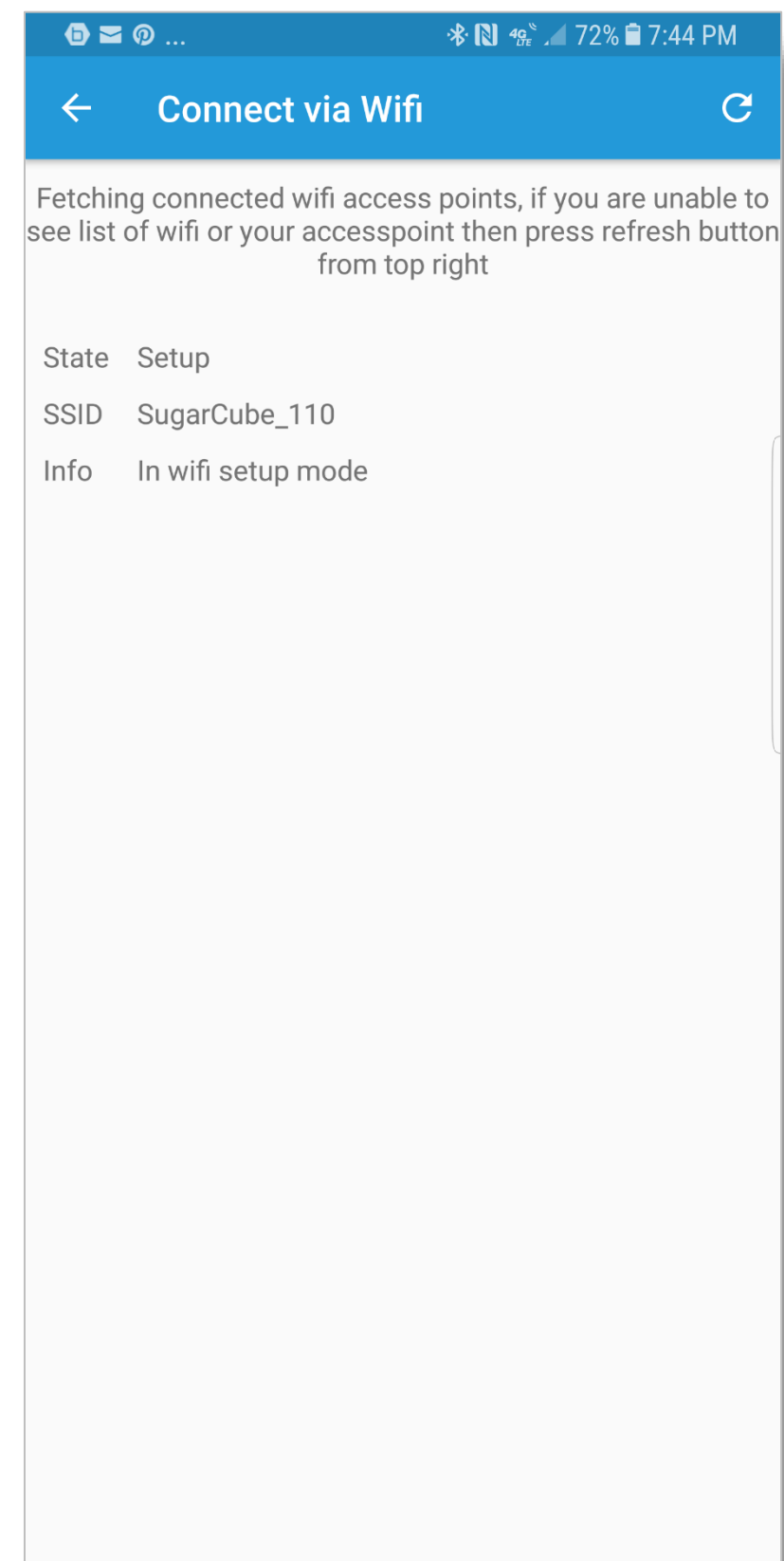


(4) User taps 'WiFi Setup' to start the process.

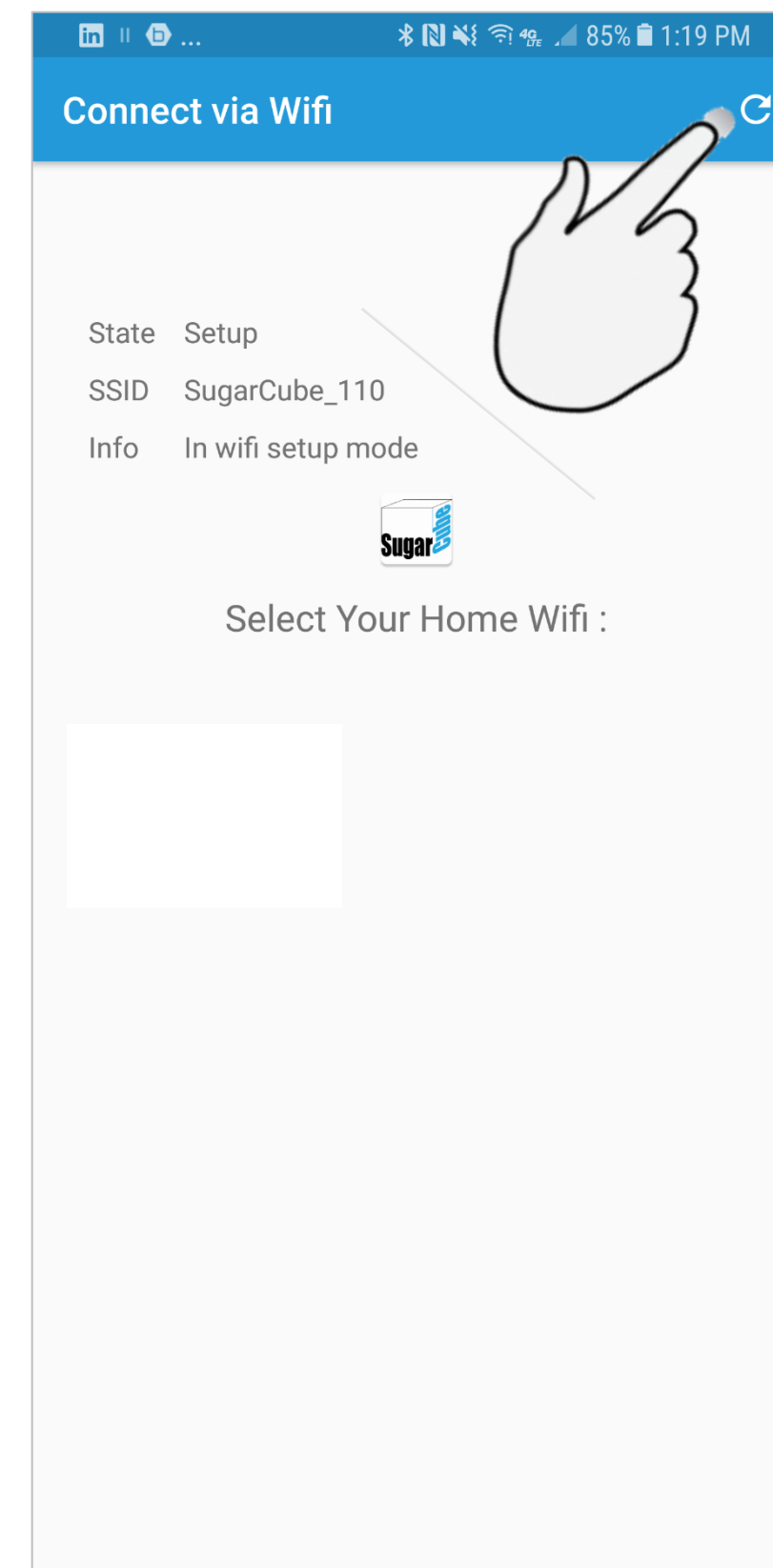
23. Connecting the app with SugarCube on WiFi



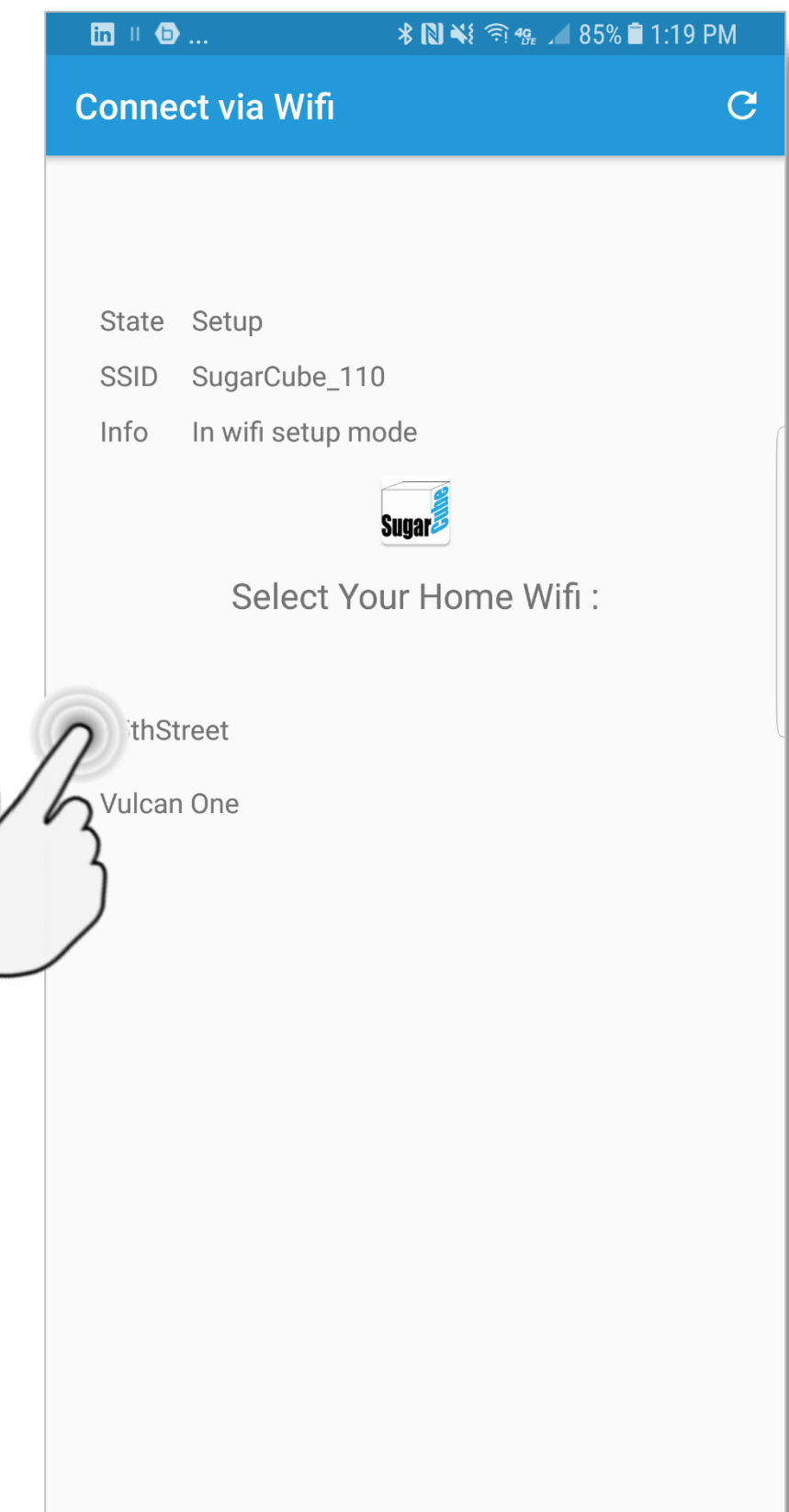
(5) Next, press and hold the pairing button (on the rear panel) for 5 seconds as instructed.



(6) After this, your SugarCube will be configured as an access point and display its SSID (e.g. SugarCube_110). In this screen the app has now connected to it automatically. The password is: **SugarCube**

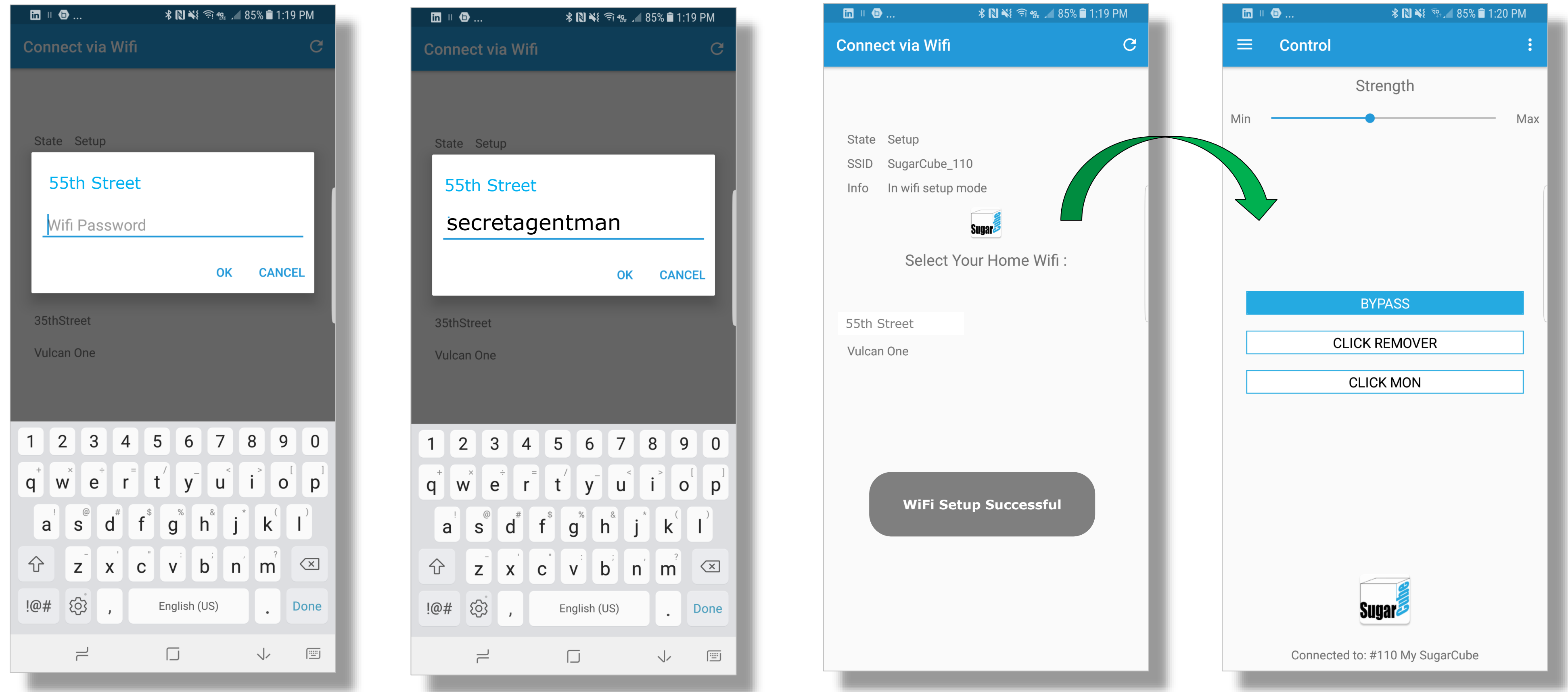


(7) Next, the SugarCube will find your available WiFi connections. This could take a minute but if nothing shows, please press the refresh (upper right).



(8) When you see your network, select the desired connection.

24. Connecting the app with SugarCube on Wi-Fi



(9) Next, enter the password for your WiFi network.

(10) The app has connected successfully with the SugarCube and will now jump to the main screen.

(11) The connection process is complete and you are ready to go.

NOTE: as of version 2.1 software, you can enter **5123** without needing to get a pair code.



Using **Wired Ethernet** with SugarCube

25. Connecting Your SC2-Mini/Phono via Wired Ethernet



If you would like to control your SC2-Mini/Phono over wired Ethernet, plug in the cable to the back panel as seen above and follow the steps on the next few pages.

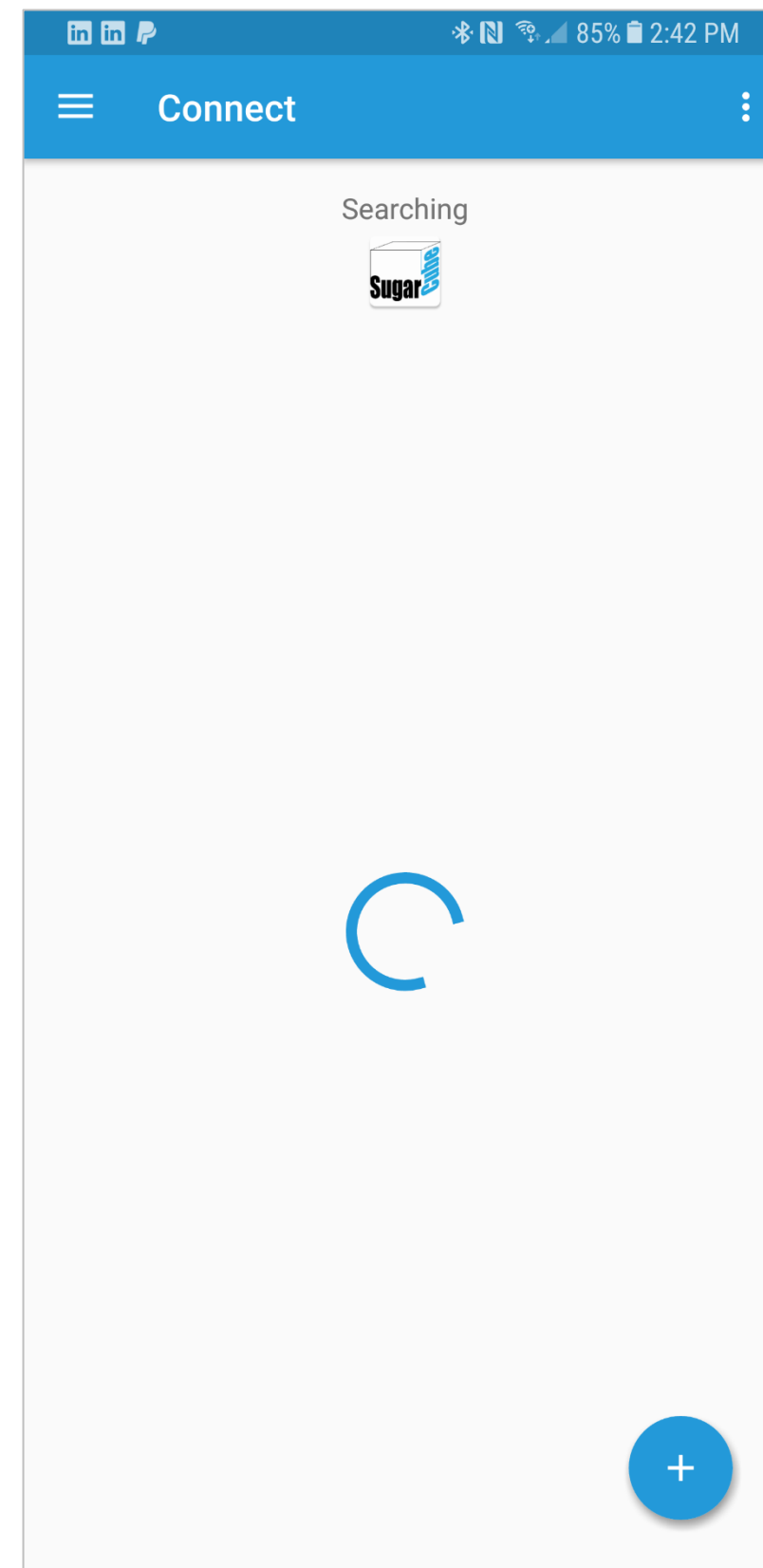
NOTE: THE SC2-Mini/Phono's basic functions of click and pop removal are fully available via front panel controls. However, the recording functions, other than start and stop recording, are only available via a table or laptop run a Google Chrome browser.

Apple iOS & Android

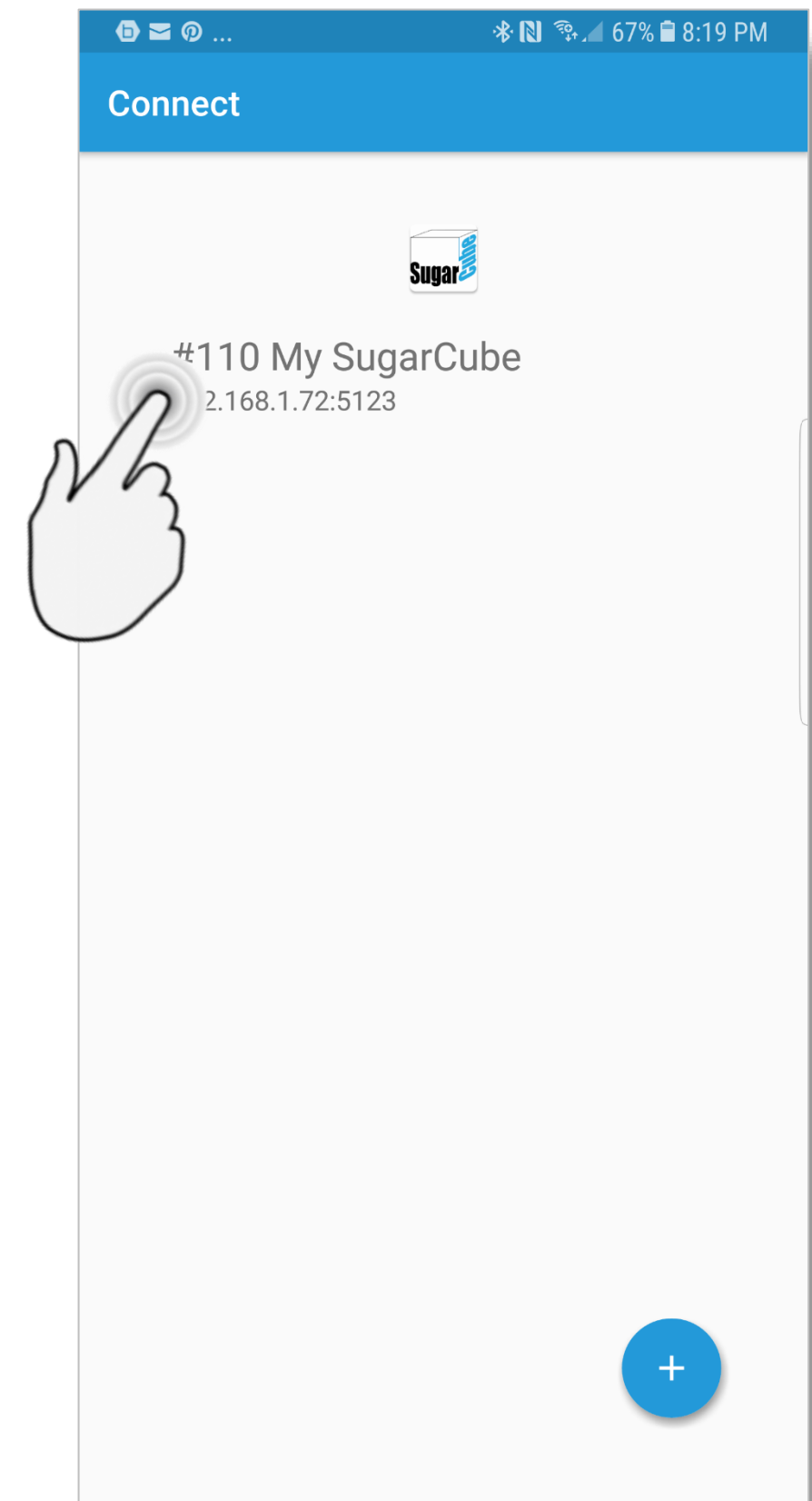
26. Connecting the app on wired Ethernet



(1) App is loading

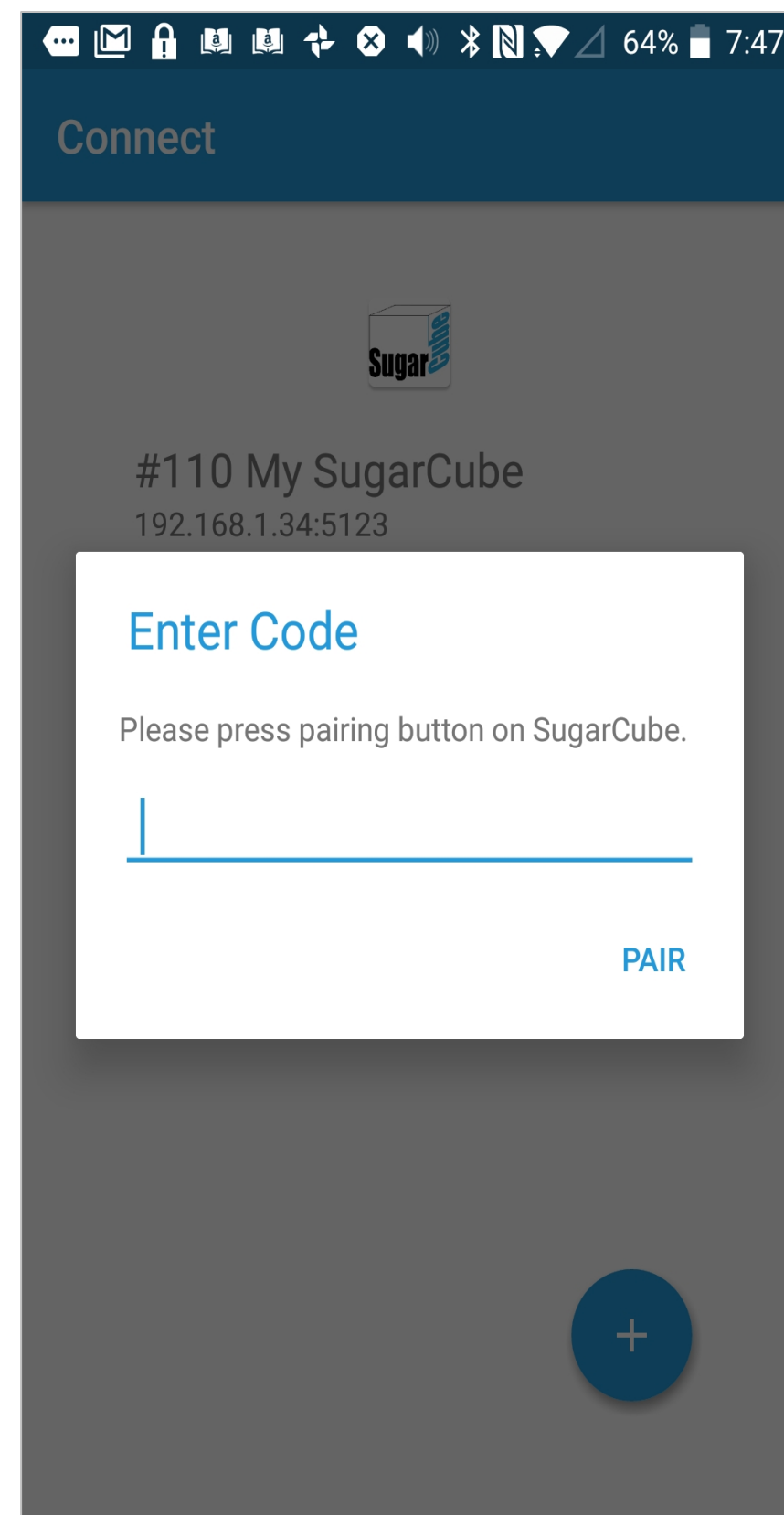


(2) App is searching for SugarCubes on the LAN

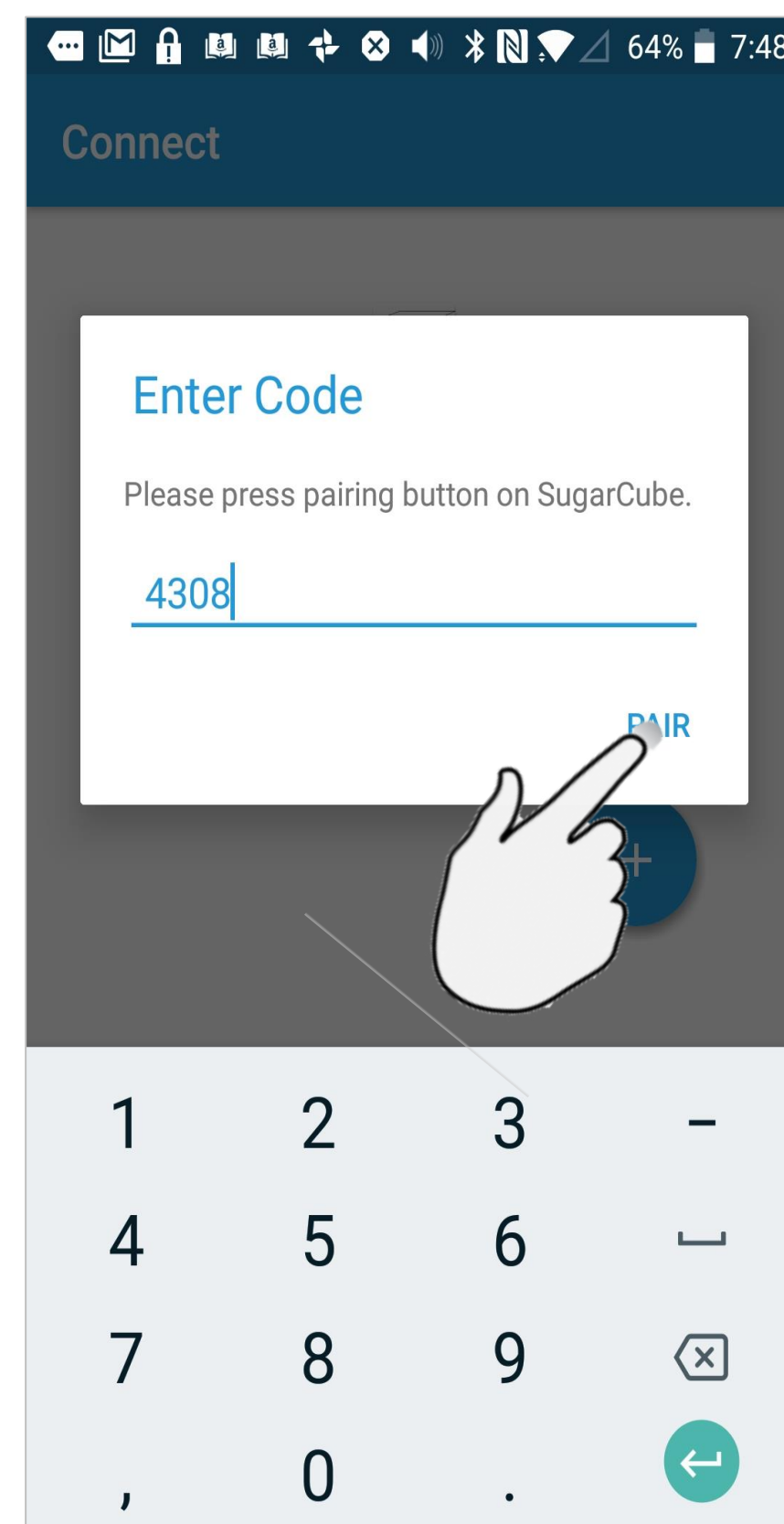


(3) App has found a SugarCube and will now initiate a connection.

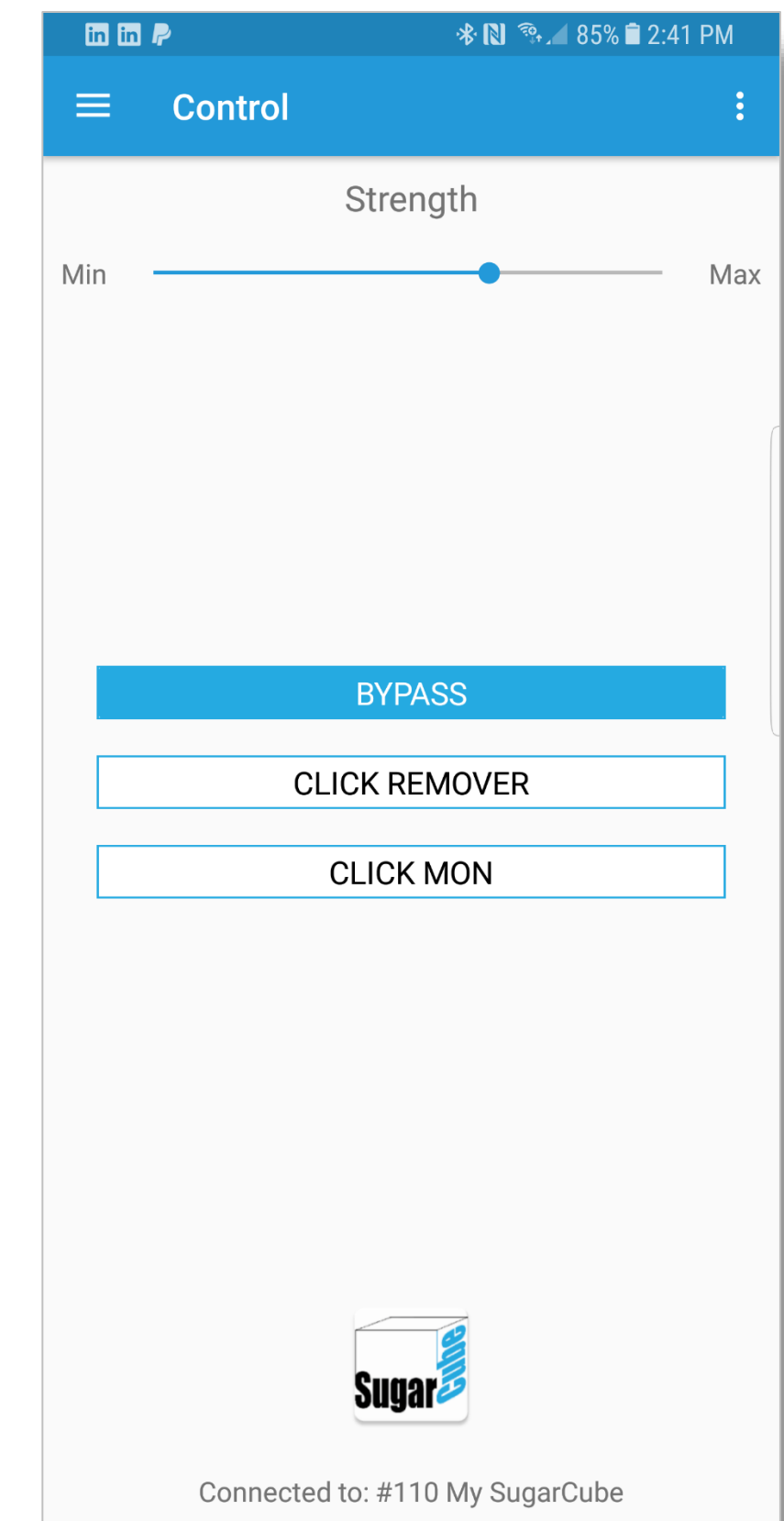
27. Connecting the app on wired Ethernet



(4) App needs you to press the knob on the front panel.



(5) App needs you to enter the pair code displayed on the front panel display then touch 'PAIR'.



(6) App is linked to SugarCube and ready for use.

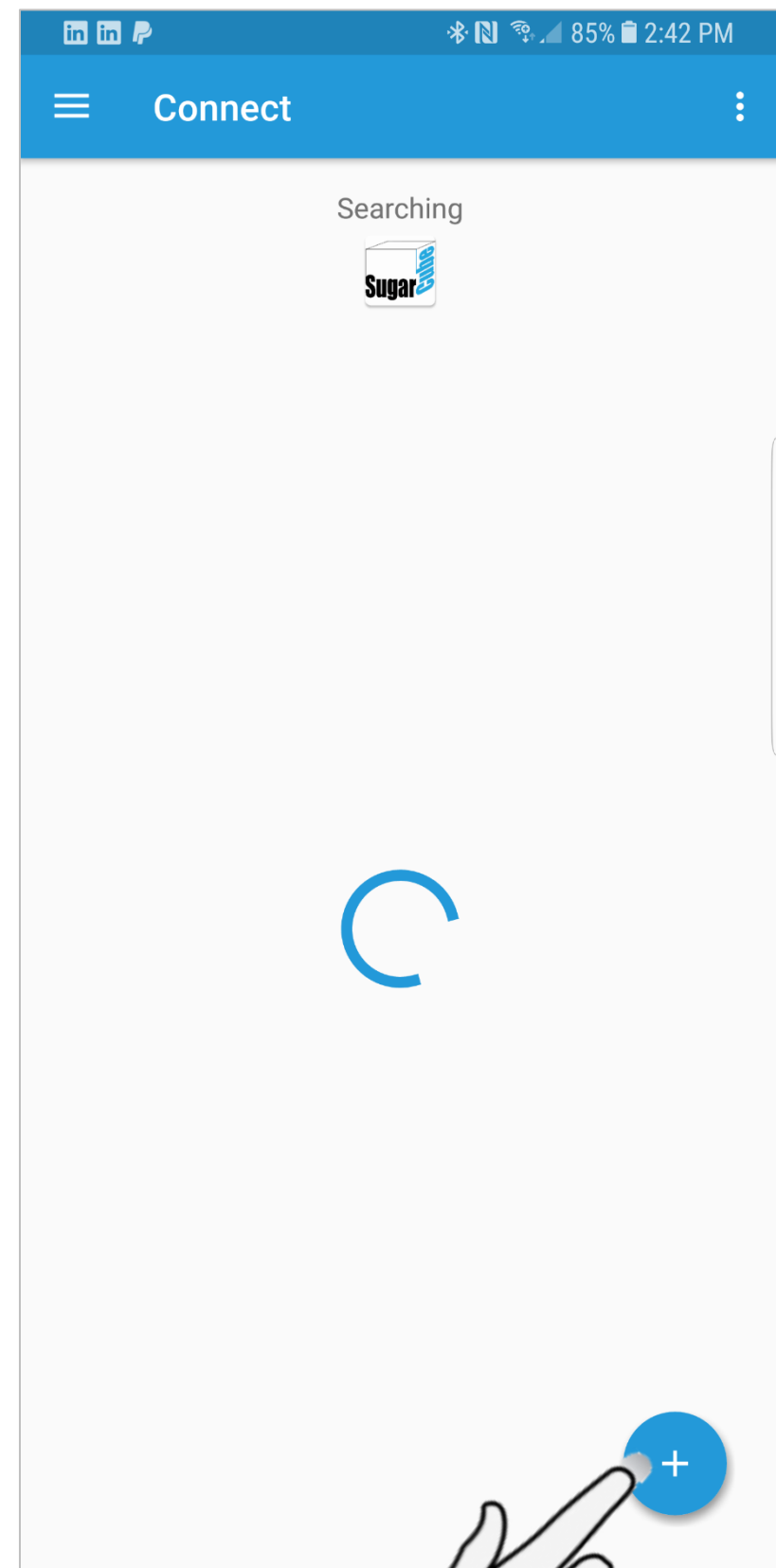
NOTE: as of version 2.1 software, you can enter **5123** without needing to get a pair code.



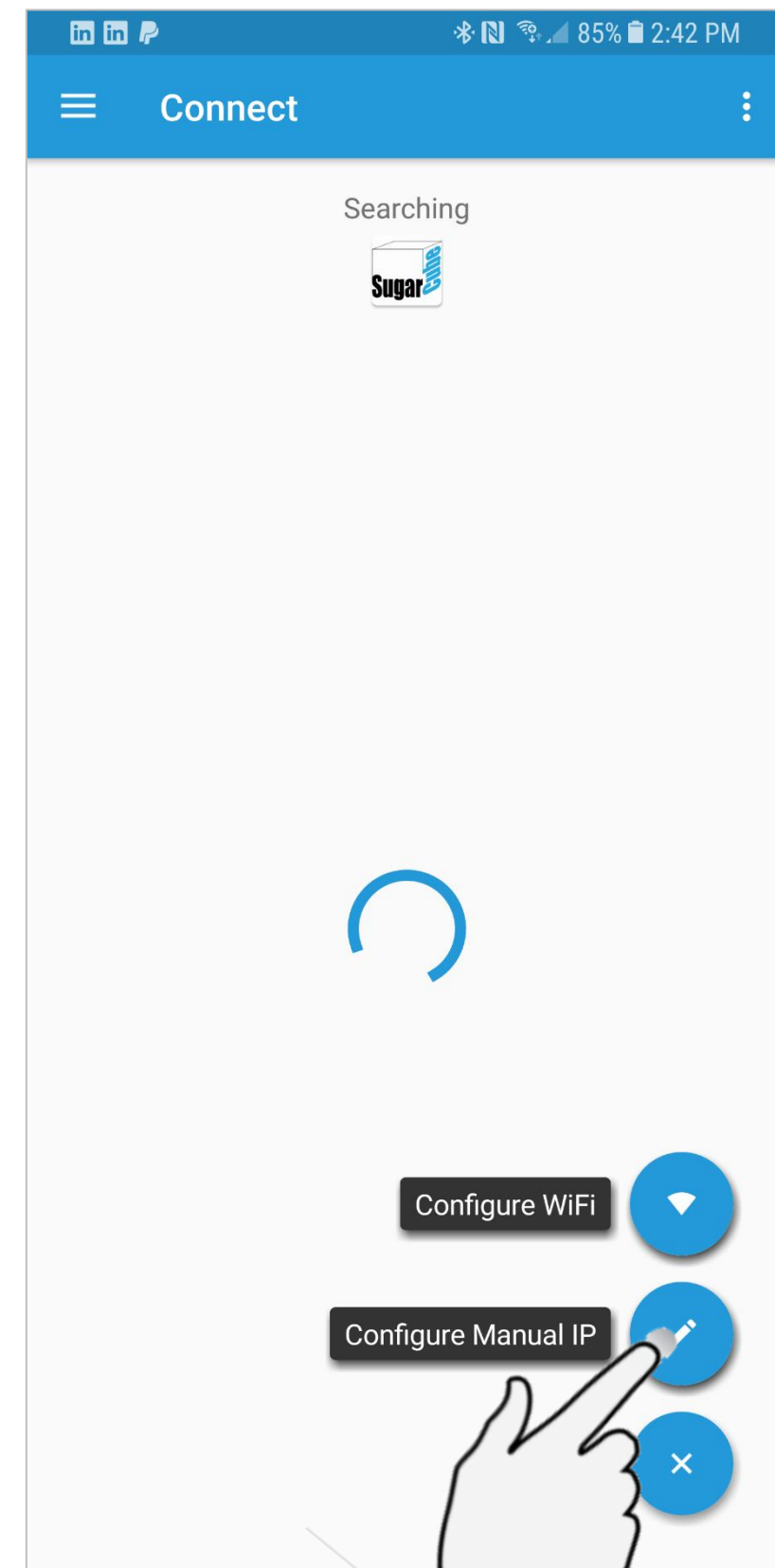
Manually Connecting to SugarCube

Using the SugarCube's IP Address

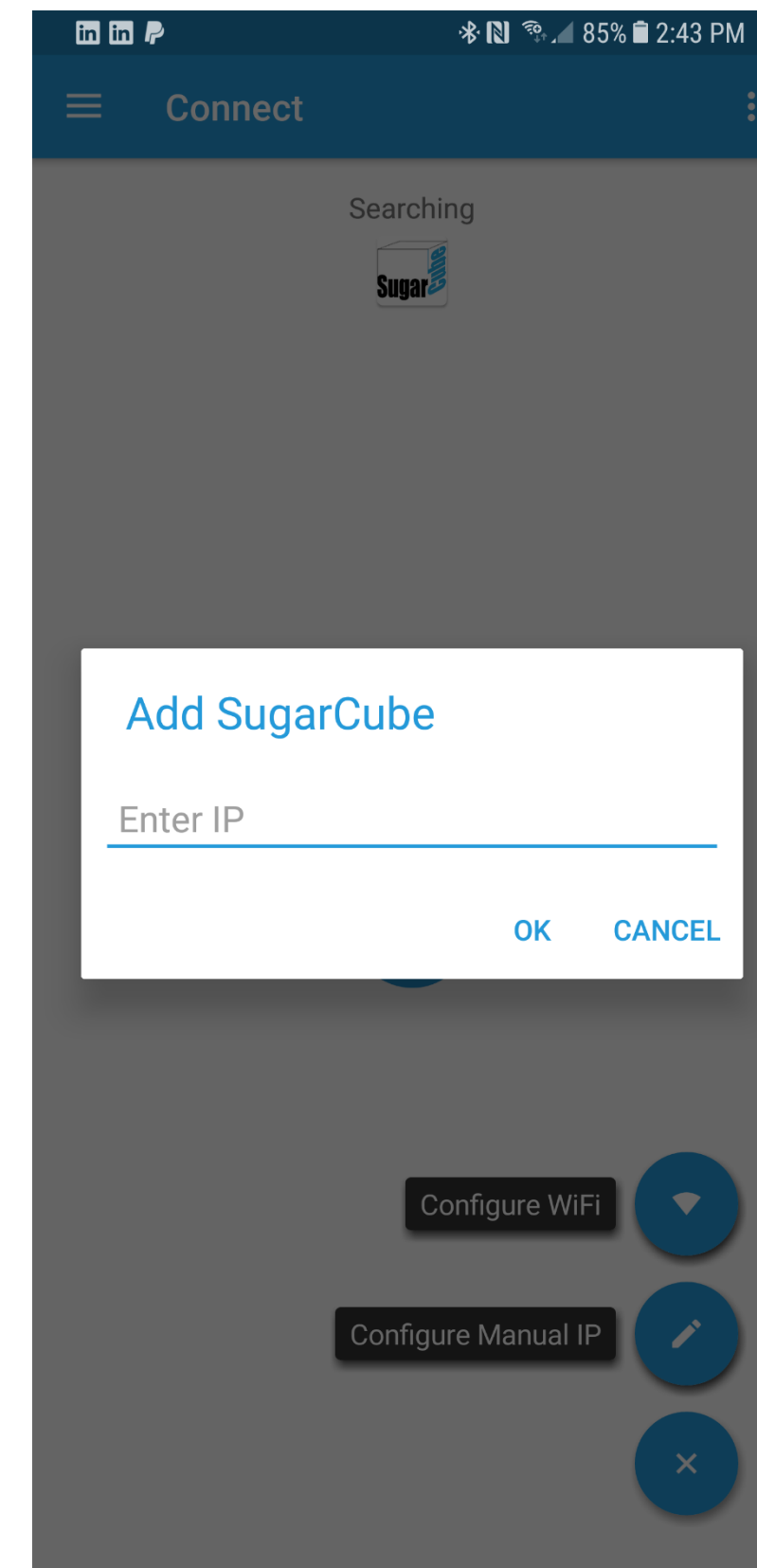
28. Manually Connecting the App Using an IP Address



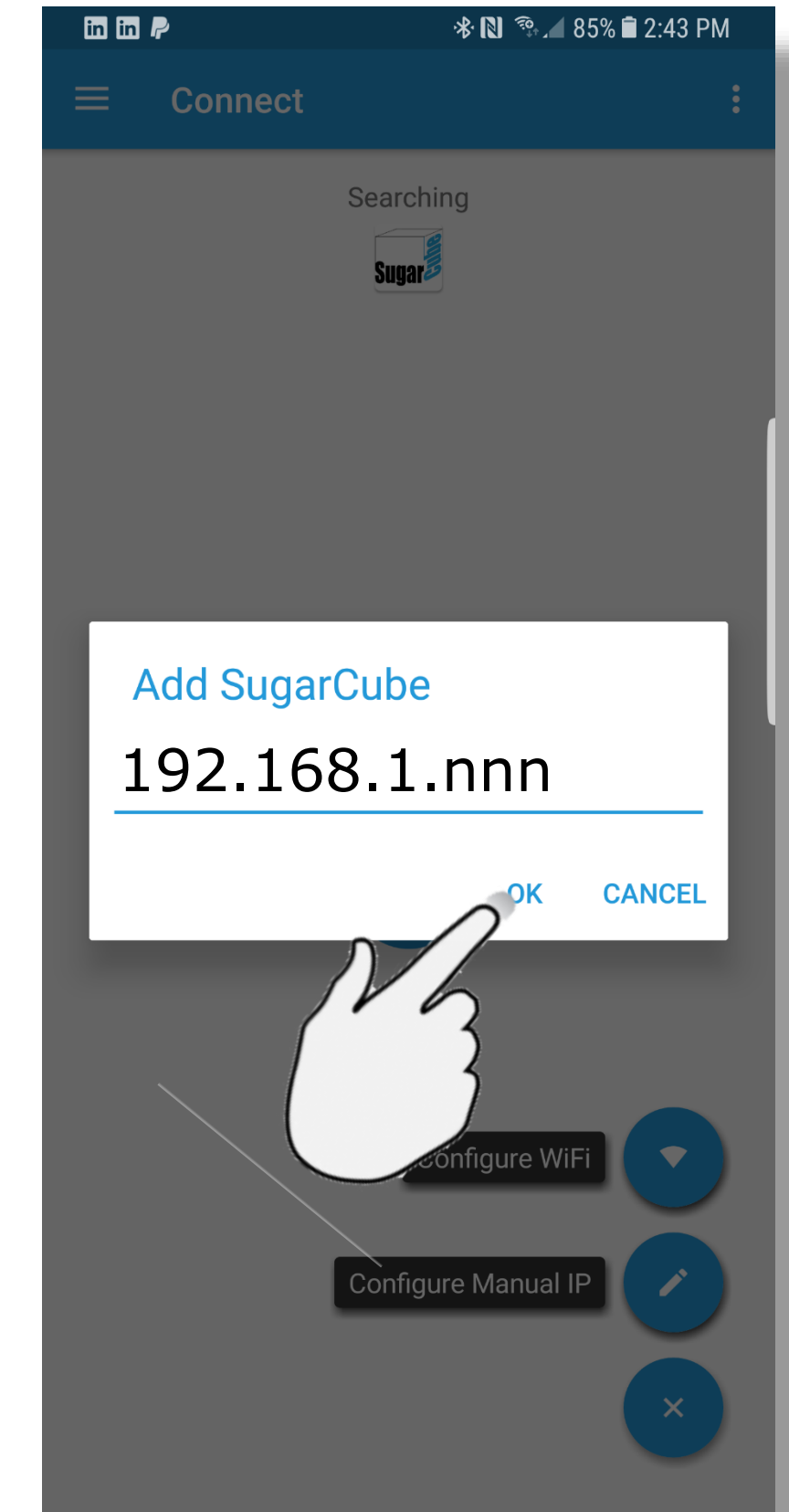
(1) If, for any reason, you wish to manually connect to your SugarCube touch the menu button in the lower right of the screen.



(2) Select 'Configure Manual IP'.

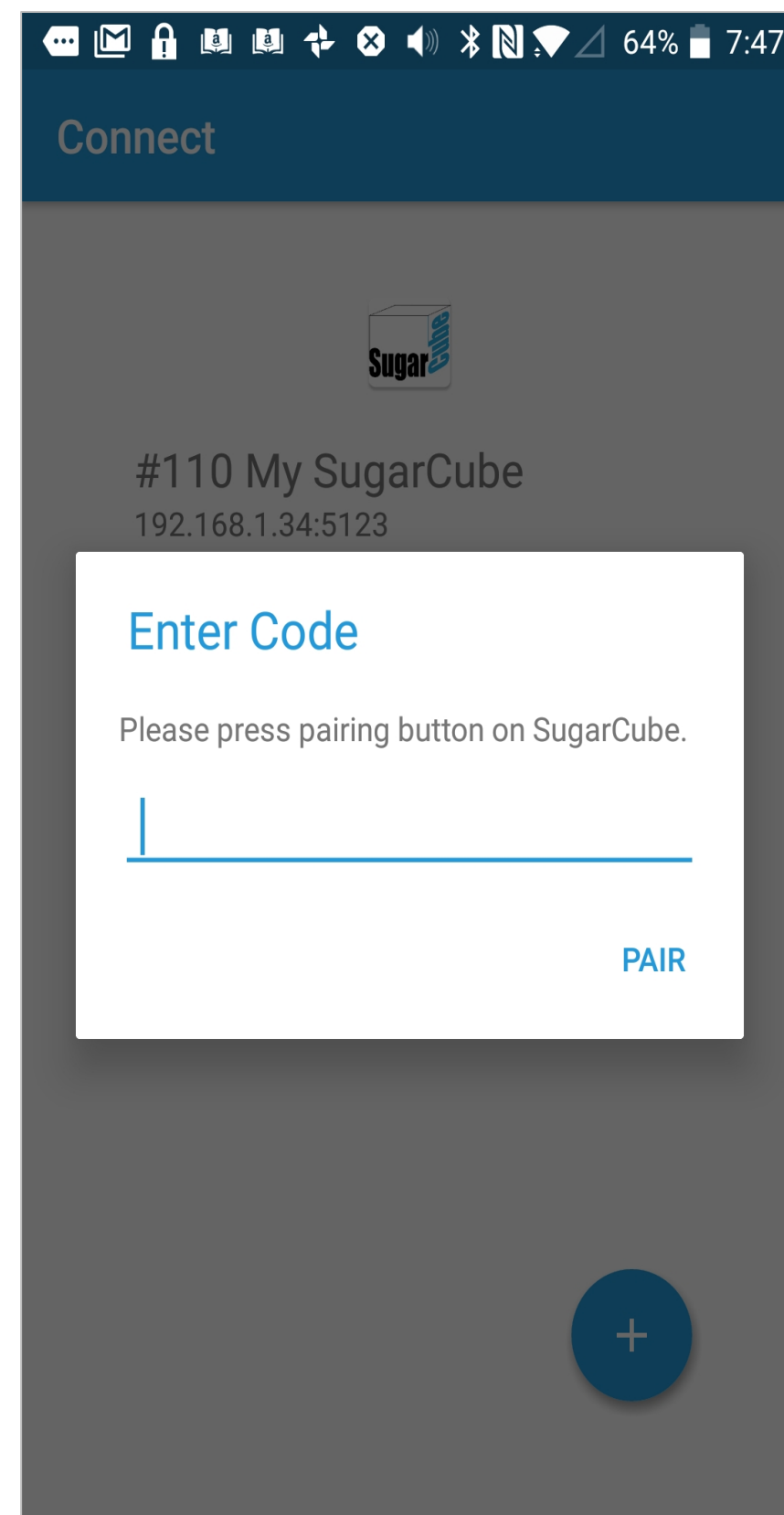


(3) You will then see a window to enter the IP address of the SugarCube. This address can be seen by pressing and holding 'Bypass' for a few seconds.

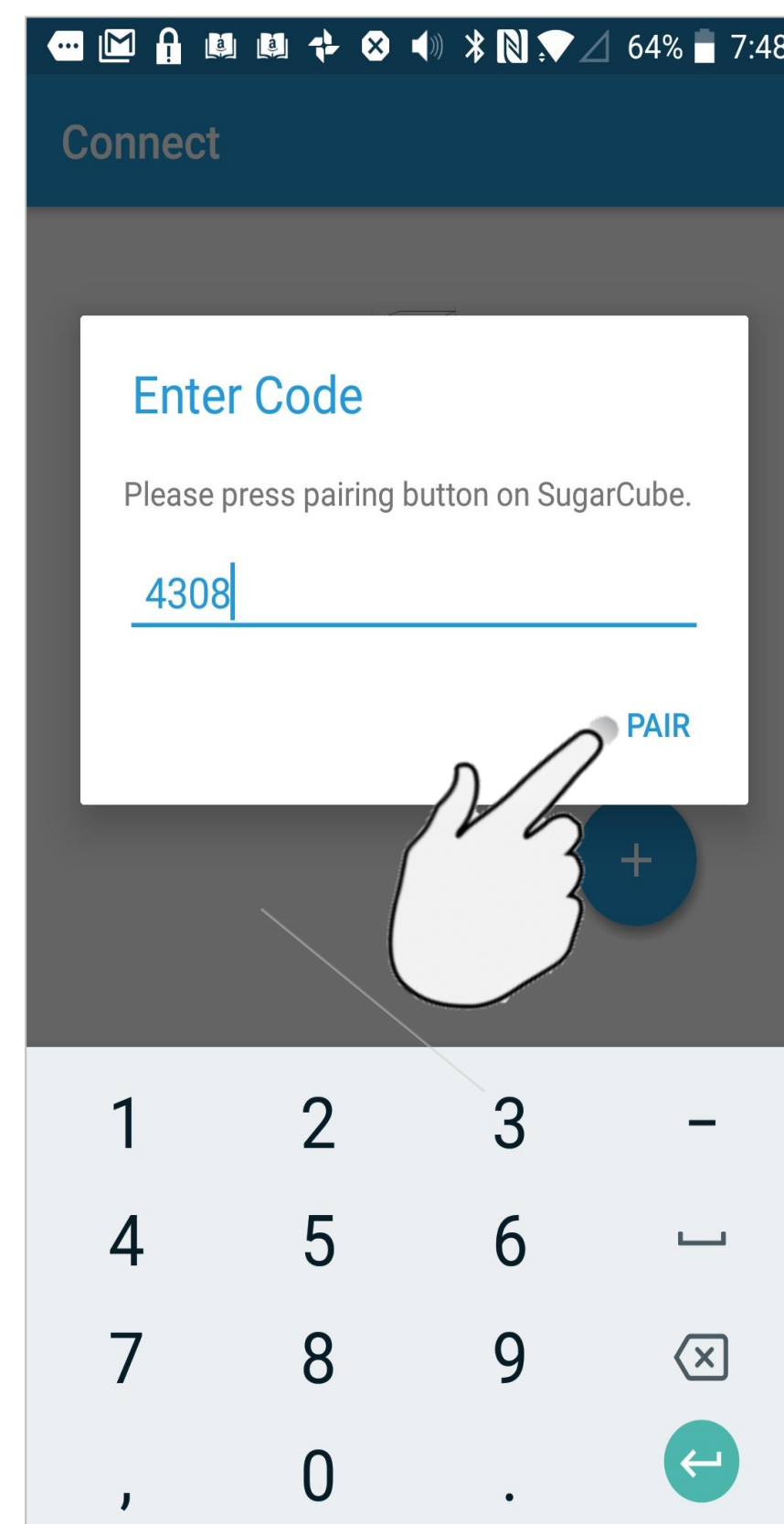


(4) Enter the IP address as displayed on the front panel display.

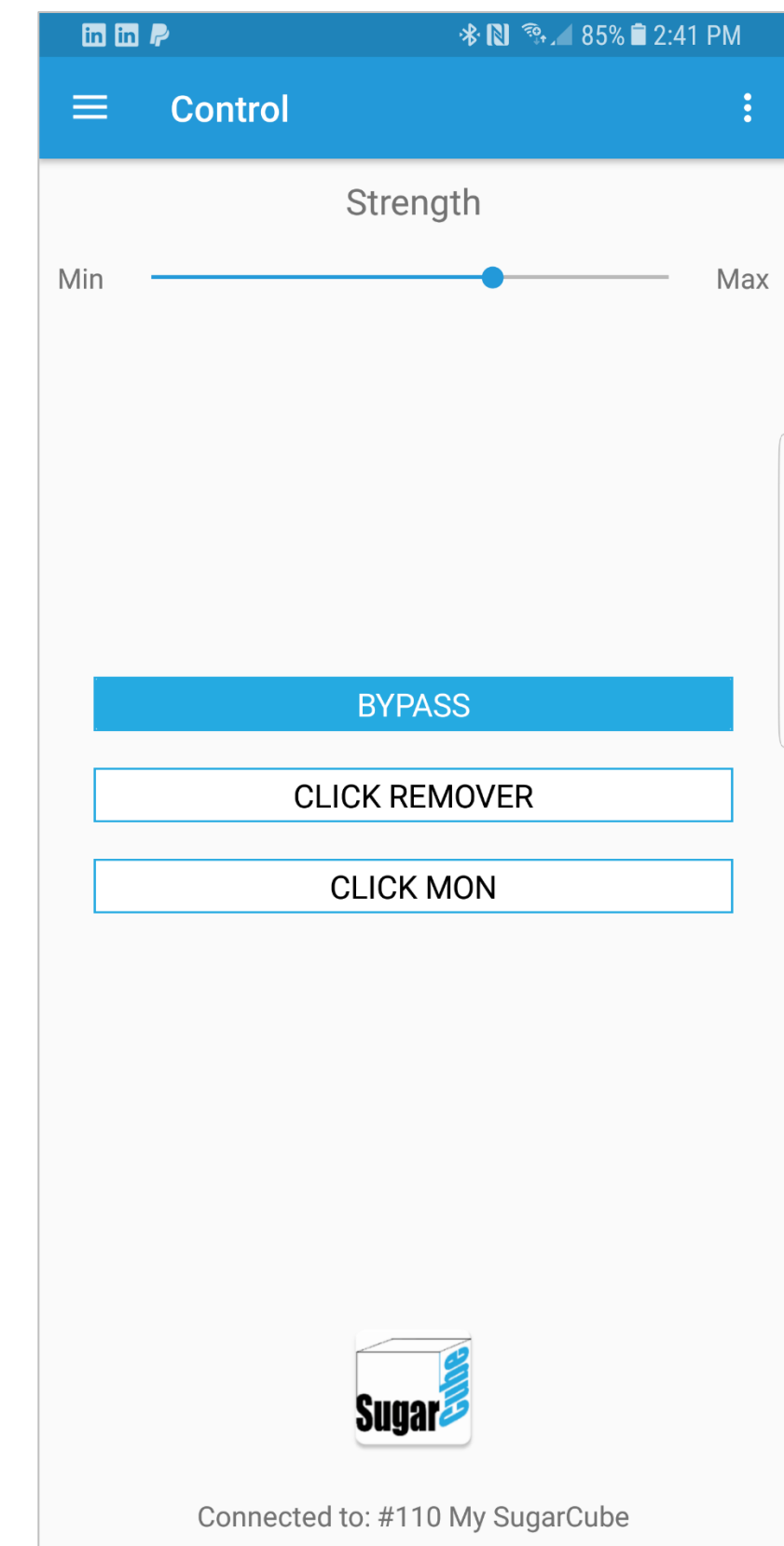
29. Manually Connecting the App Using an IP Address



(5) App need you to press the knob on the front panel.



(6) App needs you to enter the pair code displayed on the front panel display then touch 'PAIR'.



(7) App is linked to SugarCube and ready for use.

NOTE: as of version 2.1 software, you can enter **5123** without needing to get a pair code.

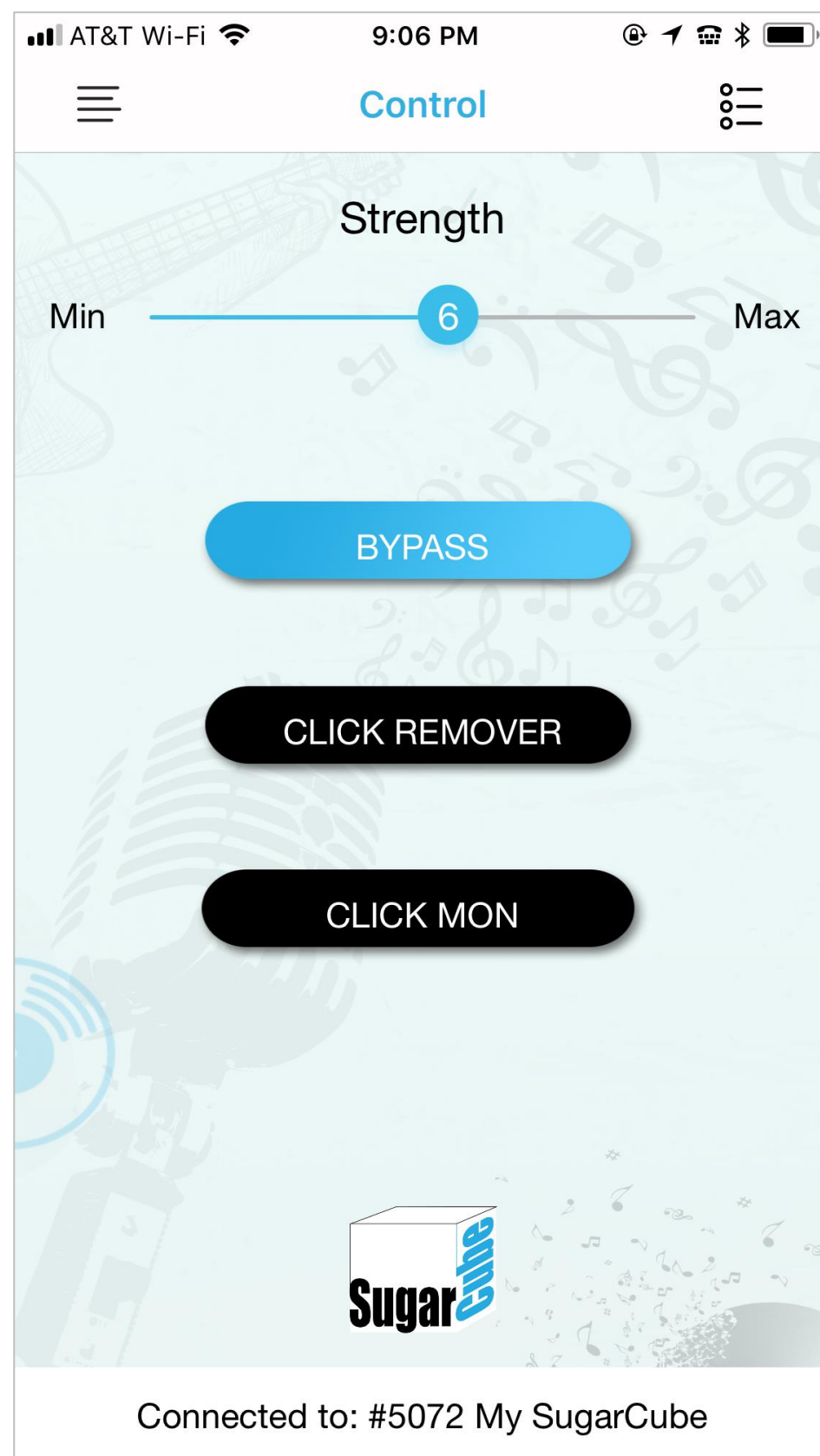
Controlling SugarCube via the App

3. Basic SugarCube Functions

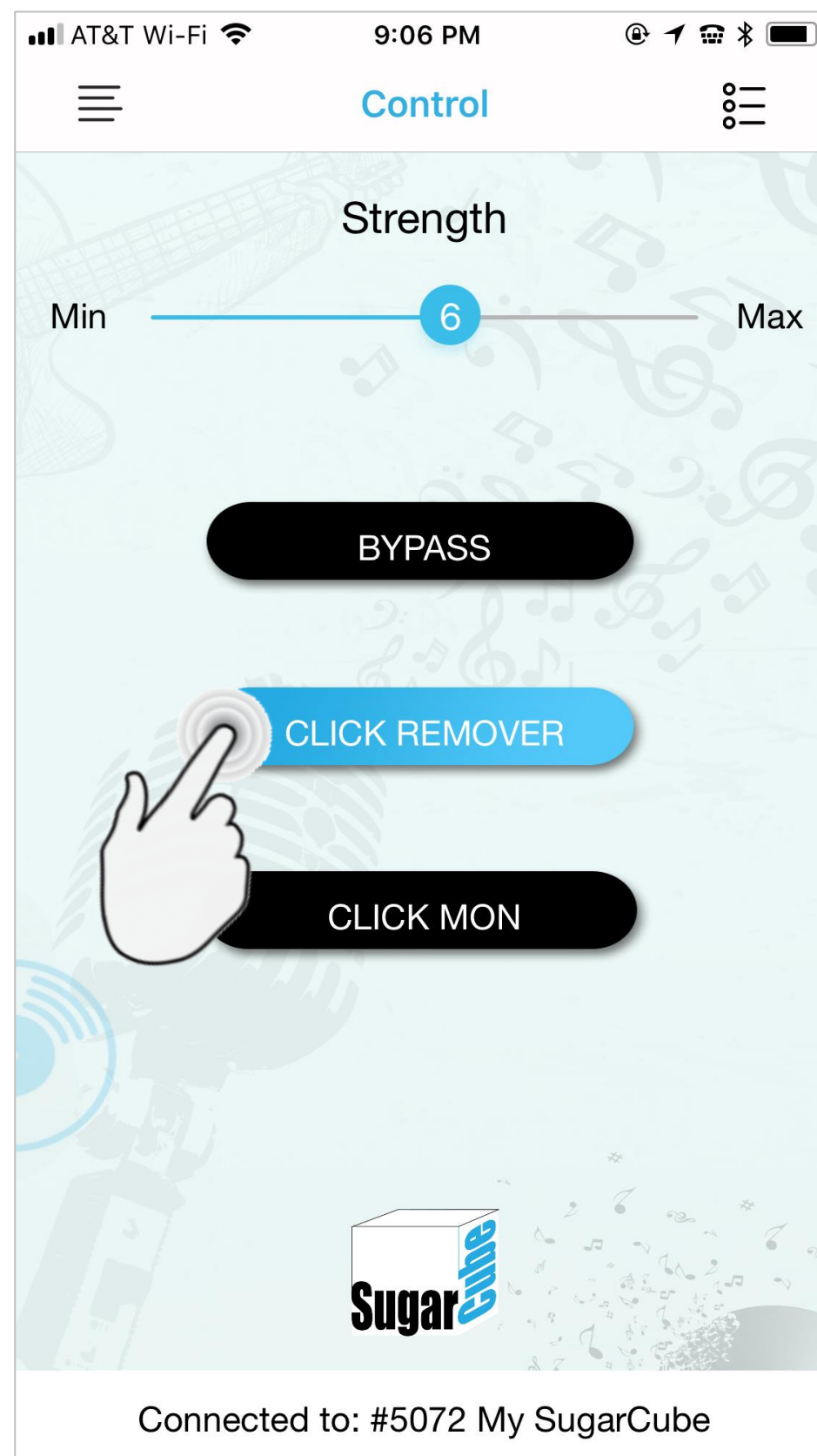
Apple iOS & Android



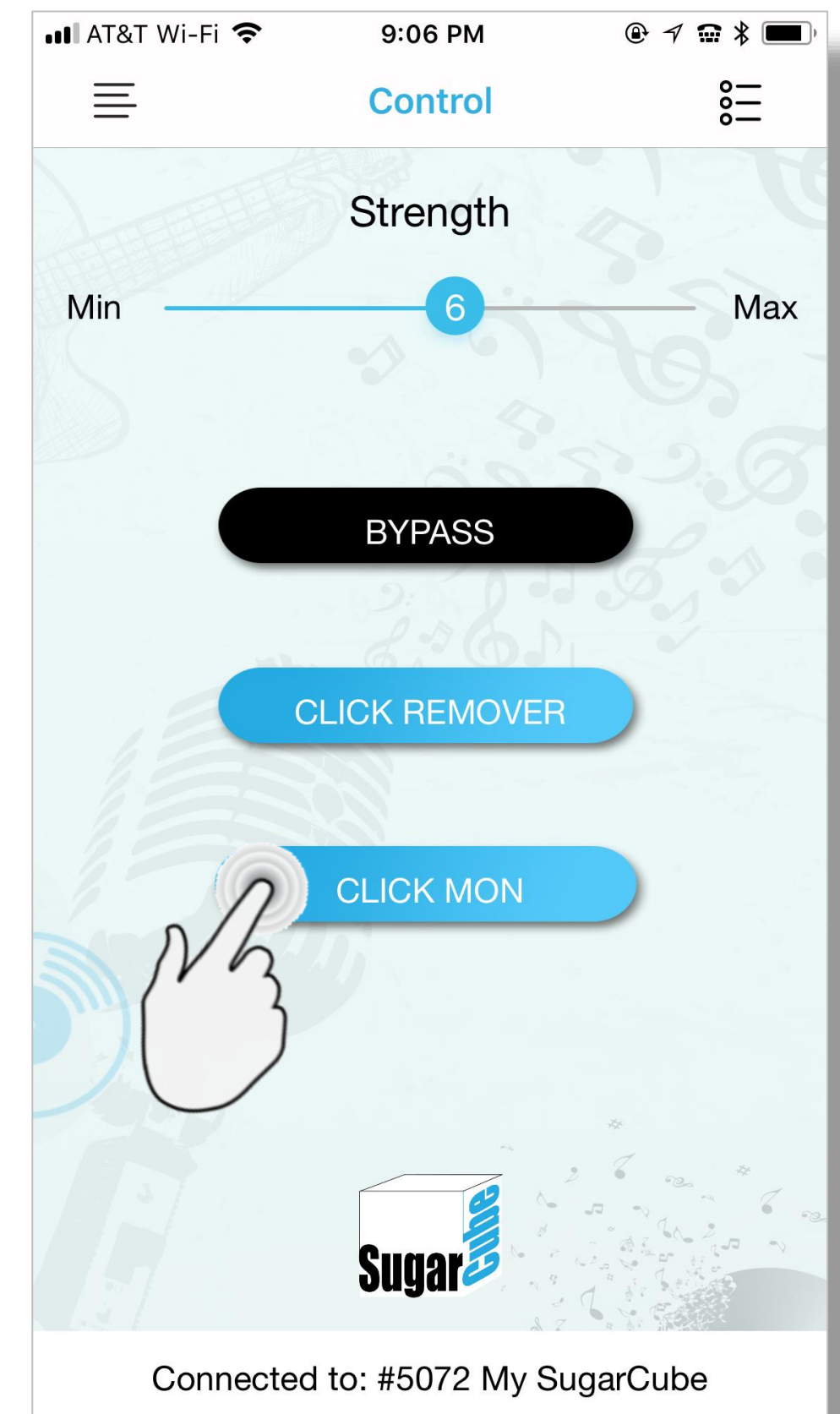
30. Main Control Page



(1) App matches the front panel controls. In this example the SugarCube is in 'Bypass' mode.

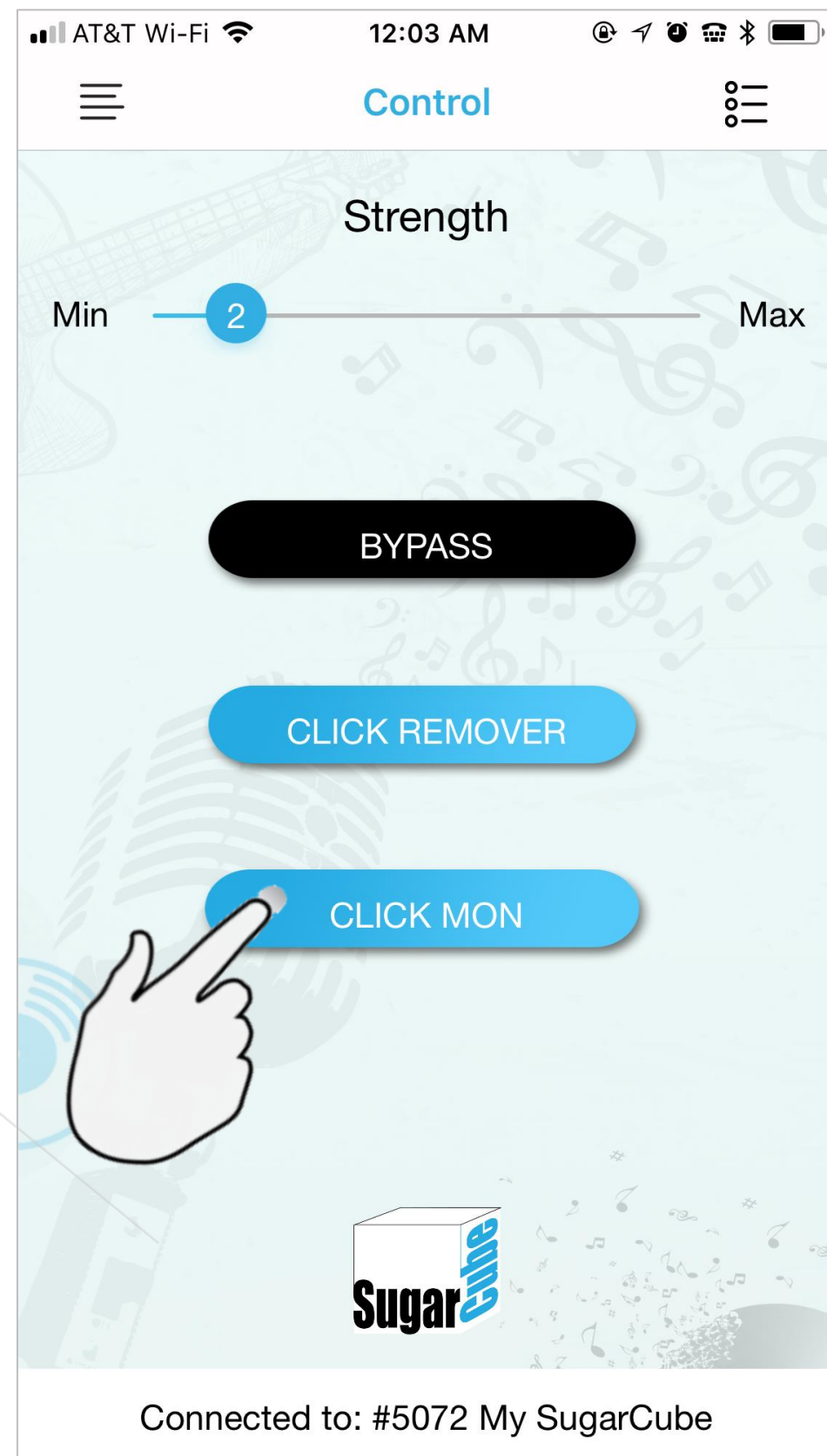


(2) You can select 'Click Remover' to engage the click removal process, same as pushing 'In' on the front panel.

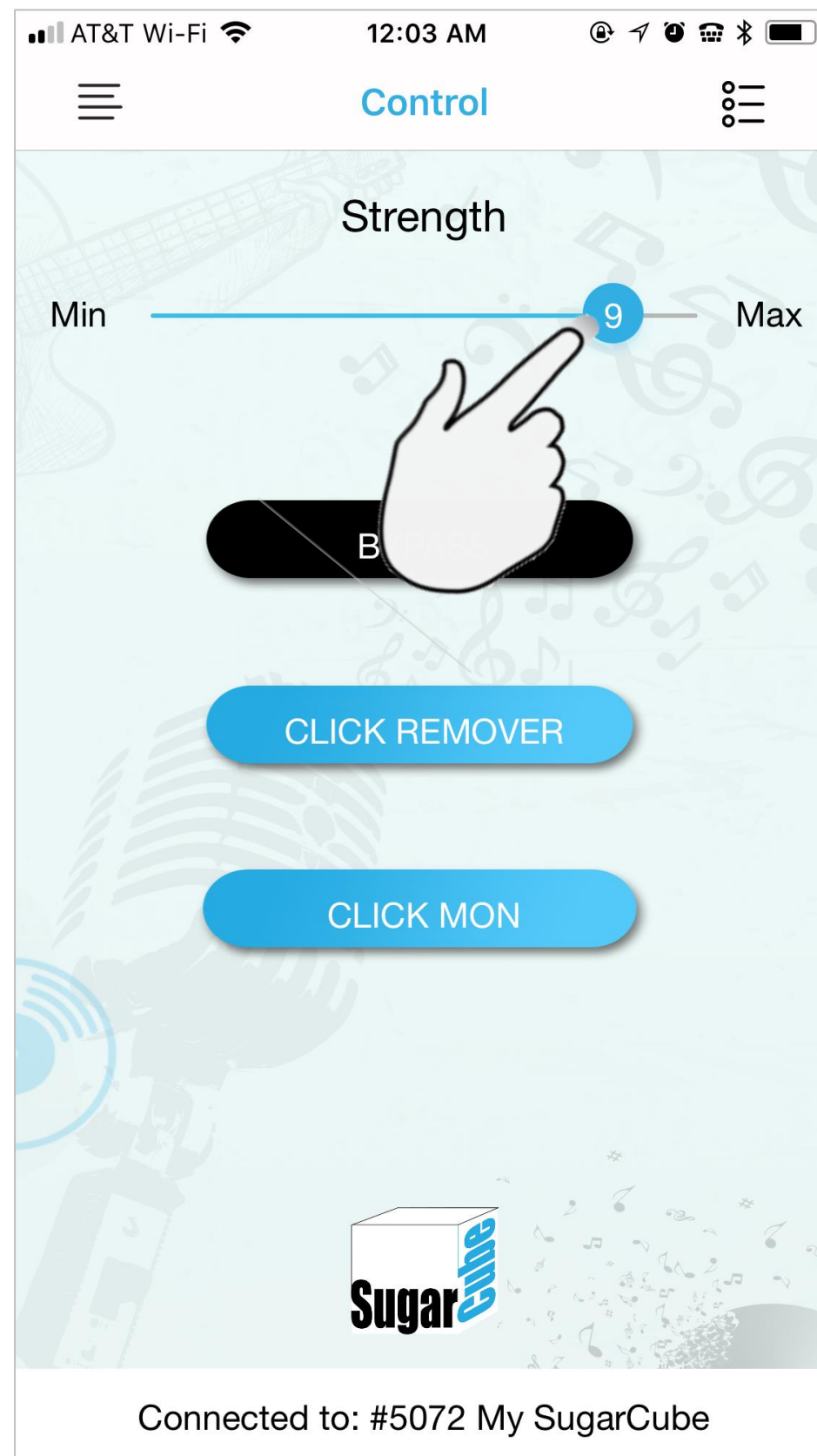


(3) You can select 'Click Mon' to listen to the level of clicks and pops while the LP plays. In this mode the music is suppressed to better hear the noise.

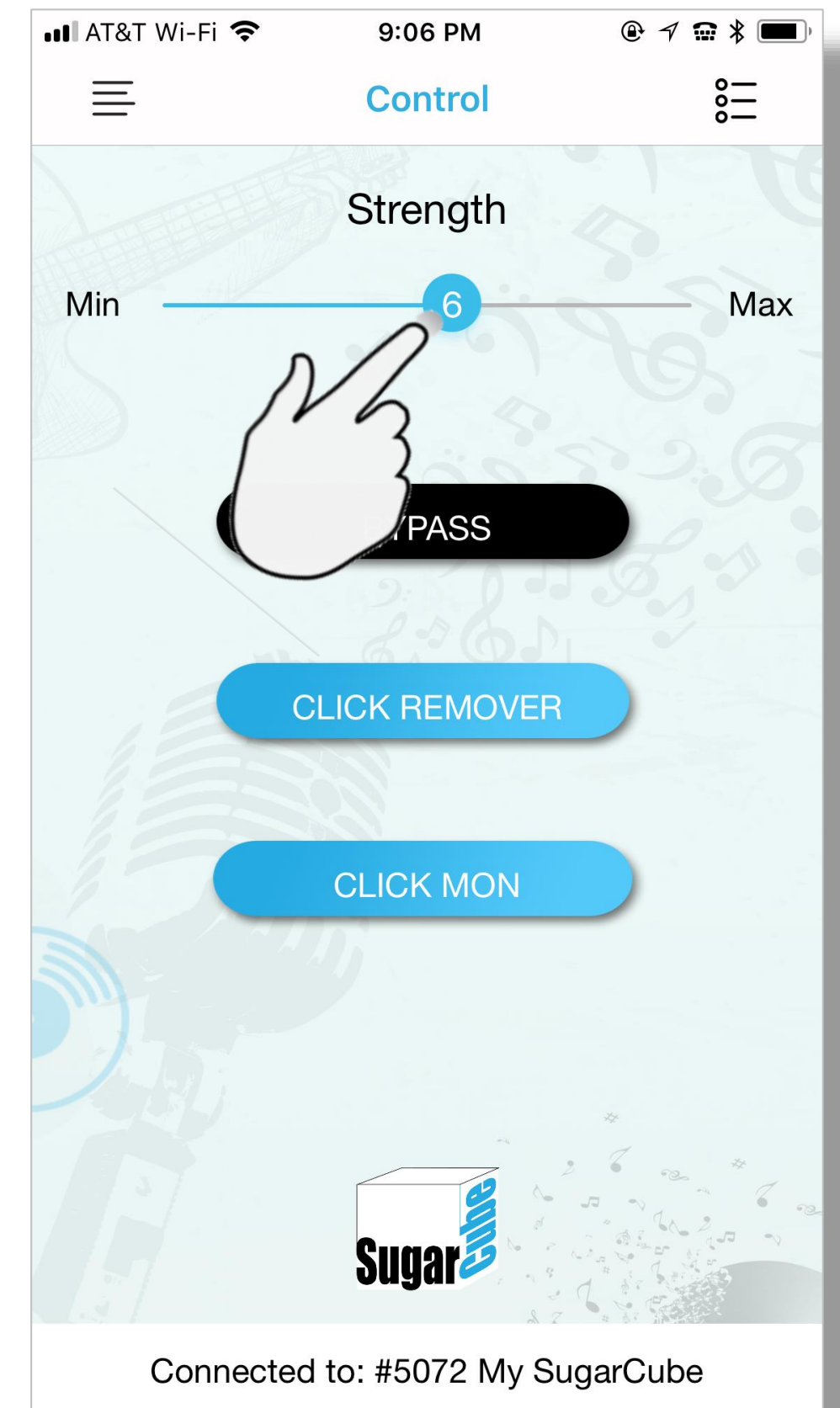
31. Setting Click Removal Strength



(4) With app or front panel set to 'Click Monitor'...



(5) You can adjust the slider to listen to the result of setting 'Strength'.



(6) You will generally find that a setting of '5' or '6' is all you need for all but the most damaged LPs.

Controlling SugarCube via a Web Browser

4. Google Chrome

MacOS, Windows
Linux, iOS, or Android



32. Controlling the SugarCube from a Web Browser

Finding the IP address of your SugarCube SC2-Mini/Phono



2

Press and hold 'BYPASS' for 3 to 5 seconds
Your SugarCube will then enter a display mode where it will show you its IP address, among other information.

You will use an address that looks like this example. It usually will start with 192 or with 10.

3



1

Plug in an Ethernet Cable

If you wish to use WiFi, before connecting the unit to your sound system, take the SugarCube to your router and plug in a wired Ethernet cable then consult diagram A4.

33. Controlling the SugarCube from a Web Browser

Entering the IP address with additional information into the browser

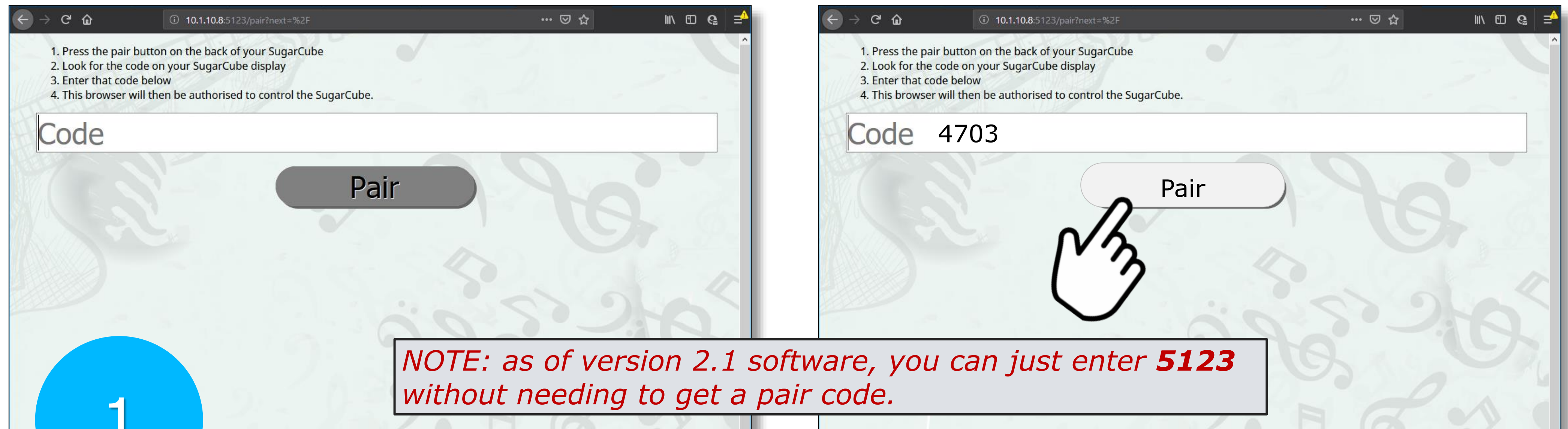


(1) Type the SugarCube's IP address into the **address bar** then add a colon ':' and '5123'.
PLEASE NOTE: the '5123' is a fixed port number and NOT the serial number of your SugarCube.

(2) Press 'Enter' and the SugarCube webpage will appear in the browser.

34. Controlling the SugarCube from a Web Browser

Pairing with the SugarCube



1



(1) You will see a screen appearing in your browser to accept a pairing code.

2

(2) Press and release the knob on the front panel.

3



(3) Enter the four-digit pairing code displayed on the SC2 display and click on 'Pair'.

35. Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 1 of 4

My SugarCube
Welcome to the new version 2.1
With improved click & pop removal
Version: 2.1-50+

Repair
Record
Settings
Support

Settings

Display
Wifi Setup
'Line In' Level
Internal / External DAC

State: normal
SSID: None
Info:
SSID
Password
Connect

Site survey Refresh
No wifi seen.

WiFi setup is where you link your SugarCube to your home WiFi if you are not using wired Ethernet to the SugarCube.

Use USB Extension Cable with WiFi adapter.

(1) If wired Ethernet is not available in your sound room, then take the SugarCube to your router just for these WiFi setup steps.

(2) You need both Ethernet and the WiFi stick plugged into the SC2 Mini/Phono.

(3) Press Site Survey to find your SSID.

36. Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 2 of 4

State: normal

Ssid: <Not Configured>

Info: disconnected

SSID

Password

Site survey

	SSID	Signal	Channel	Security	Access Points
Use	1qa2ws3ed	87%	6 40	WPA2	2
Use	219Guest	20%	11	WPA1 WPA2	1
Use	219SecureIoT	34%	6	WPA1 WPA2	1
Use	219SecureWireless	30%	6	WPA1 WPA2	1
Use	BlueCart Wifi	30%	6	WPA2	1
Use	Cart Wifi - Guest	50%	6	WPA2	1
Use	SmugMug	84%	40	WPA2	1
Use	SmugMugs	84%	40	WPA2	1
Use	sweetvinyl	100%	9 48	WPA2	4
Use	Tap Guest	34%	11	WPA1 WPA2	1

(4) Please select your network by clicking on the 'Use' button next to your SSID.

Info: disconnected

SSID

Password

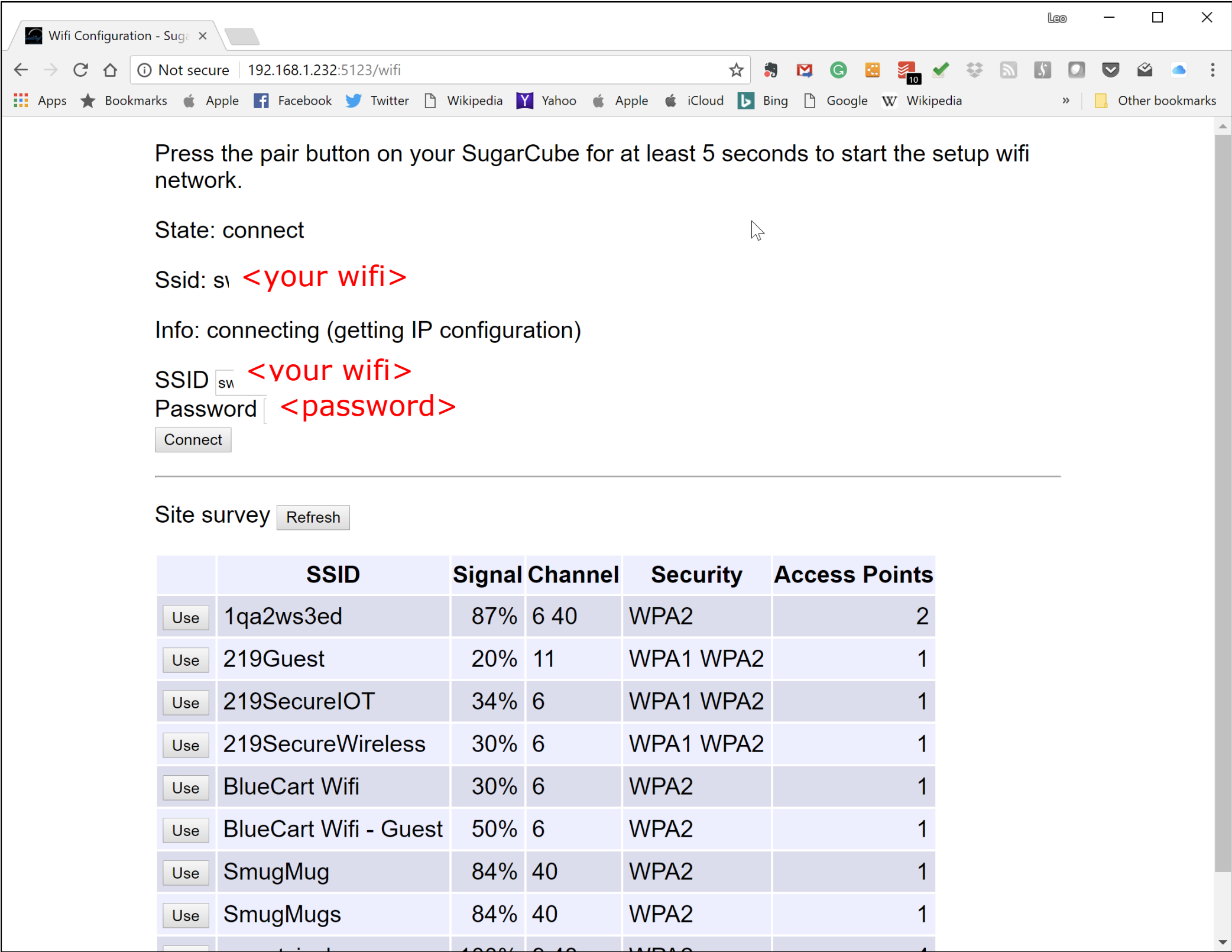
Site survey

	SSID	Signal	Channel	Security	Access
Use	1qa2ws3ed	80%	6 40	WPA2	
Use	219Guest	44%	11	WPA1 WPA2	
Use	219SecureIoT	30%	6	WPA1 WPA2	
Use	219SecureWireless	34%	6	WPA1 WPA2	
Use	BlueCart Wifi	34%	6	WPA2	
Use	SmugMug	87%	6 40	WPA2	

(5) Next, enter the password for your WiFi then click on 'Connect.'

37. Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 3 of 4



WiFi IP address

Your WiFi SSID

Wired IP address

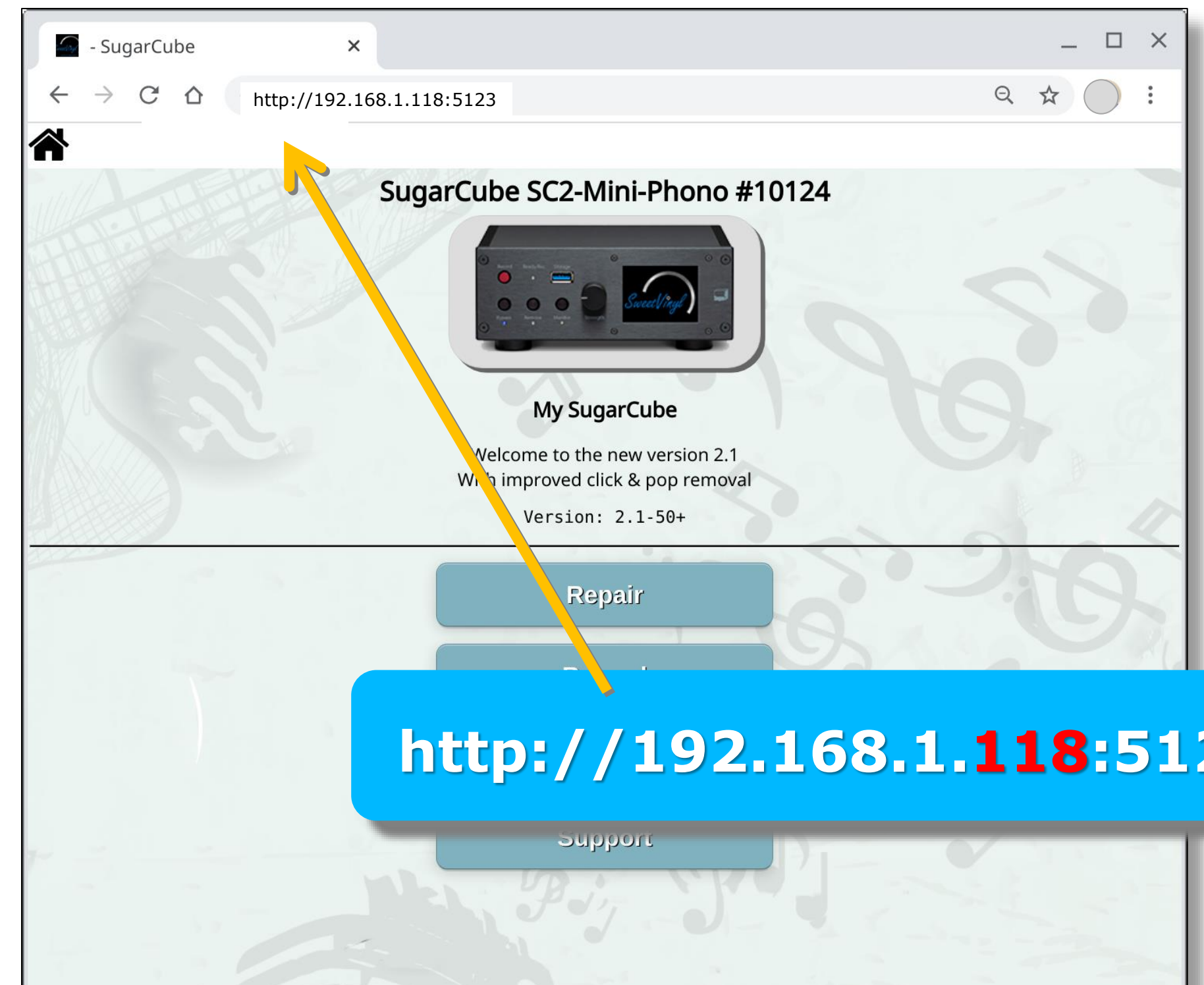


(6) The system will indicate that it is connecting to your WiFi.

(7) Once connected, you will now see, scrolling by, both your new WiFi address (w) and your currently wired Ethernet IP address (e). Remember, you are currently communicating with the SugarCube via the wired IP address.

38. Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 4 of 4



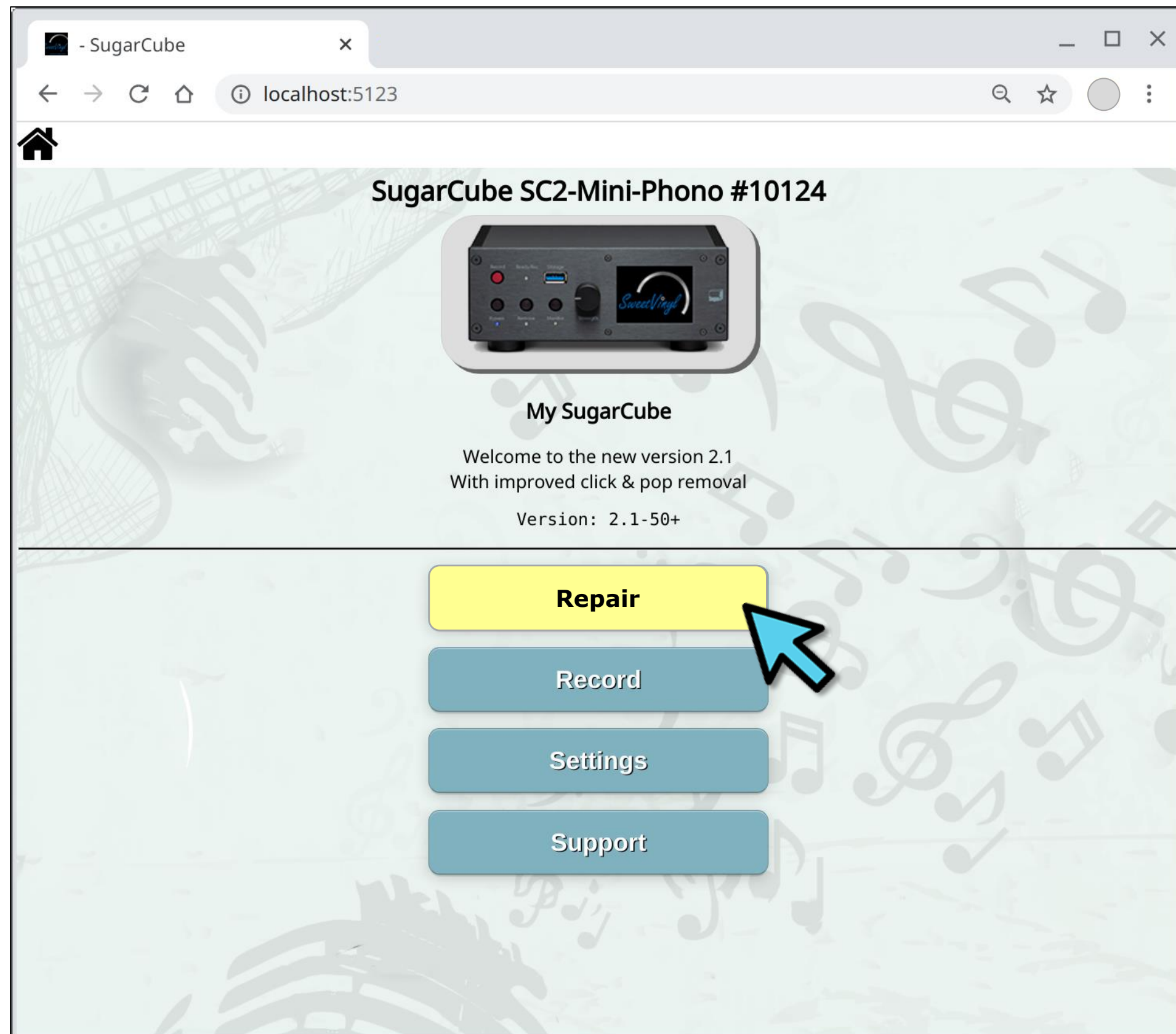
(9) So, when you disconnect the wire, your browser will no longer communicate (of course). You will have to change the IP address in your browser, in this example, to the WiFi address ending in '.118'. (Once the wire is disconnected, the front panel display will show only the WiFi IP address.)

(10) Usually, you need only to change the last group of numbers and press 'Enter', and you should reconnect (sometimes you may have to re-enter the entire address).

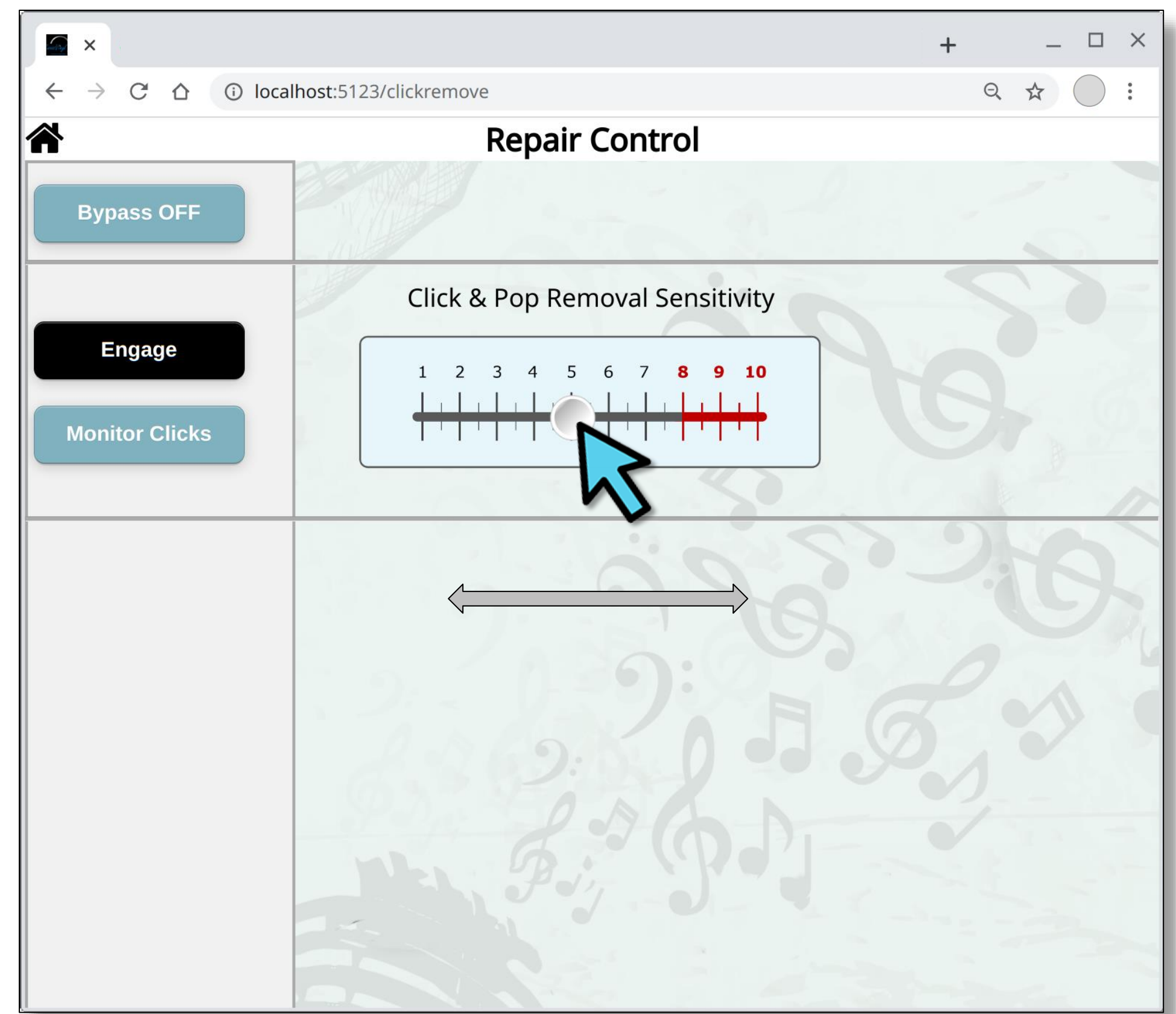
(11) Please note, if your SugarCube cannot connect to your WiFi, contact support@sweetvinyl.com with the details of your system (WiFi make and model and the serial number of your SugarCube.)

39. Controlling the SugarCube from a Web Browser

Controlling the SugarCube



(1) Click on 'Repair'



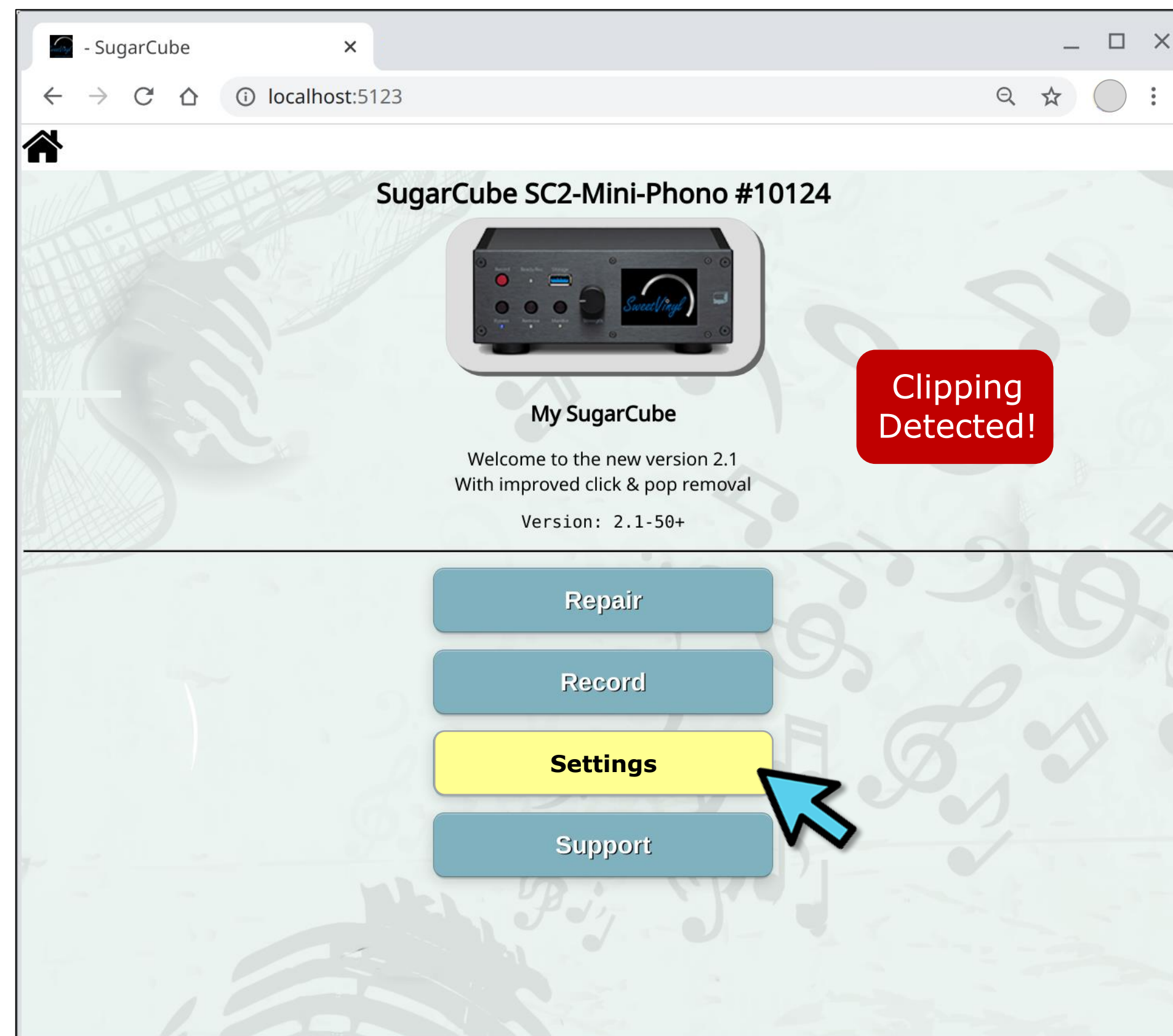
(2) Functions available mirror the front panel controls. A change here is reflected on the front panel and a change on the front panel changes the above.

40. Controlling the SugarCube from a Web Browser

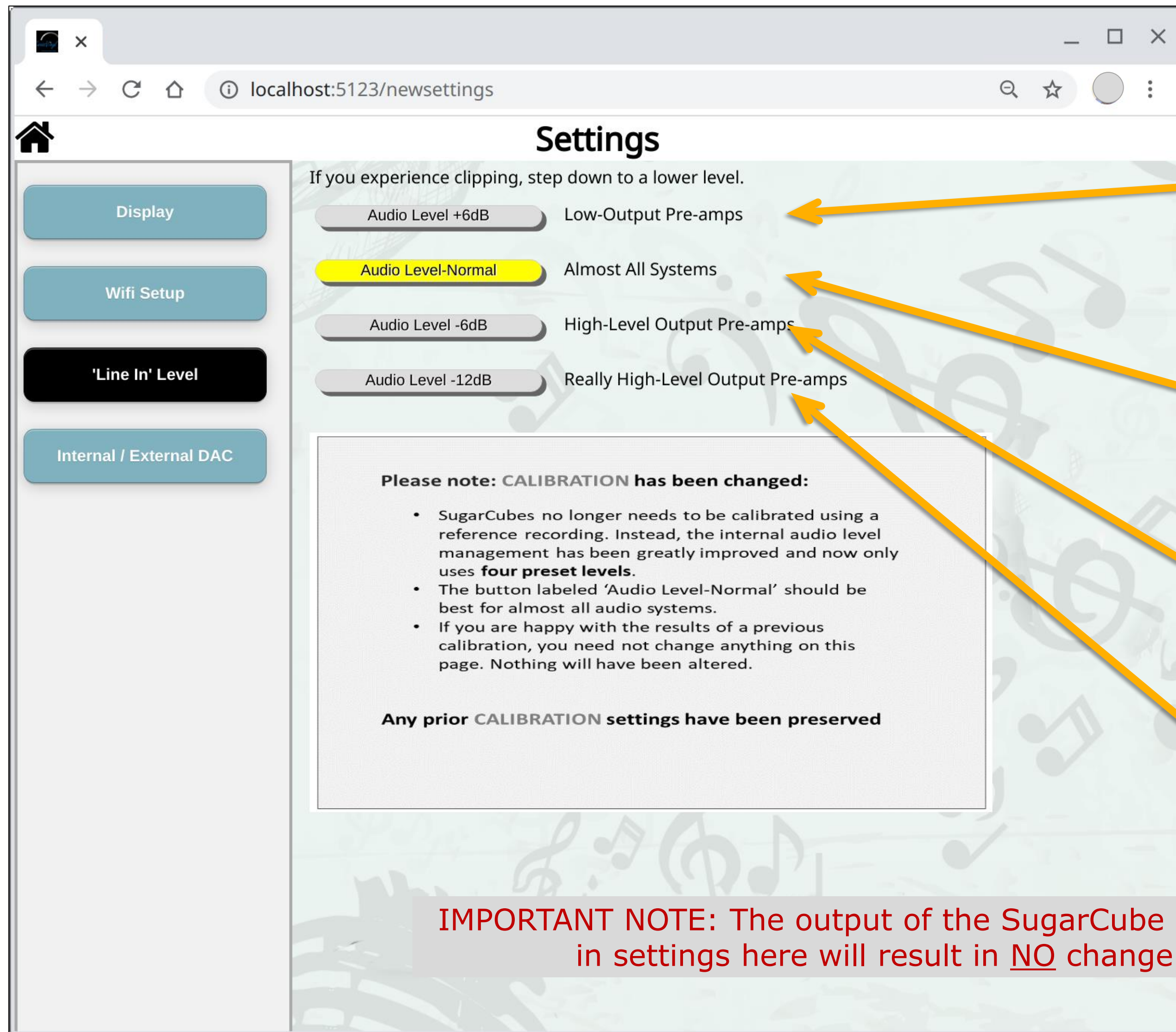
Input Level Sensitivity

PLEASE NOTE: unless you see a '**CLIPPING**' warning appear while using your SugarCube, you do **NOT** need to change the audio setup.

If you do see a clipping warning then follow the steps on the next three pages:



41. Audio Setup



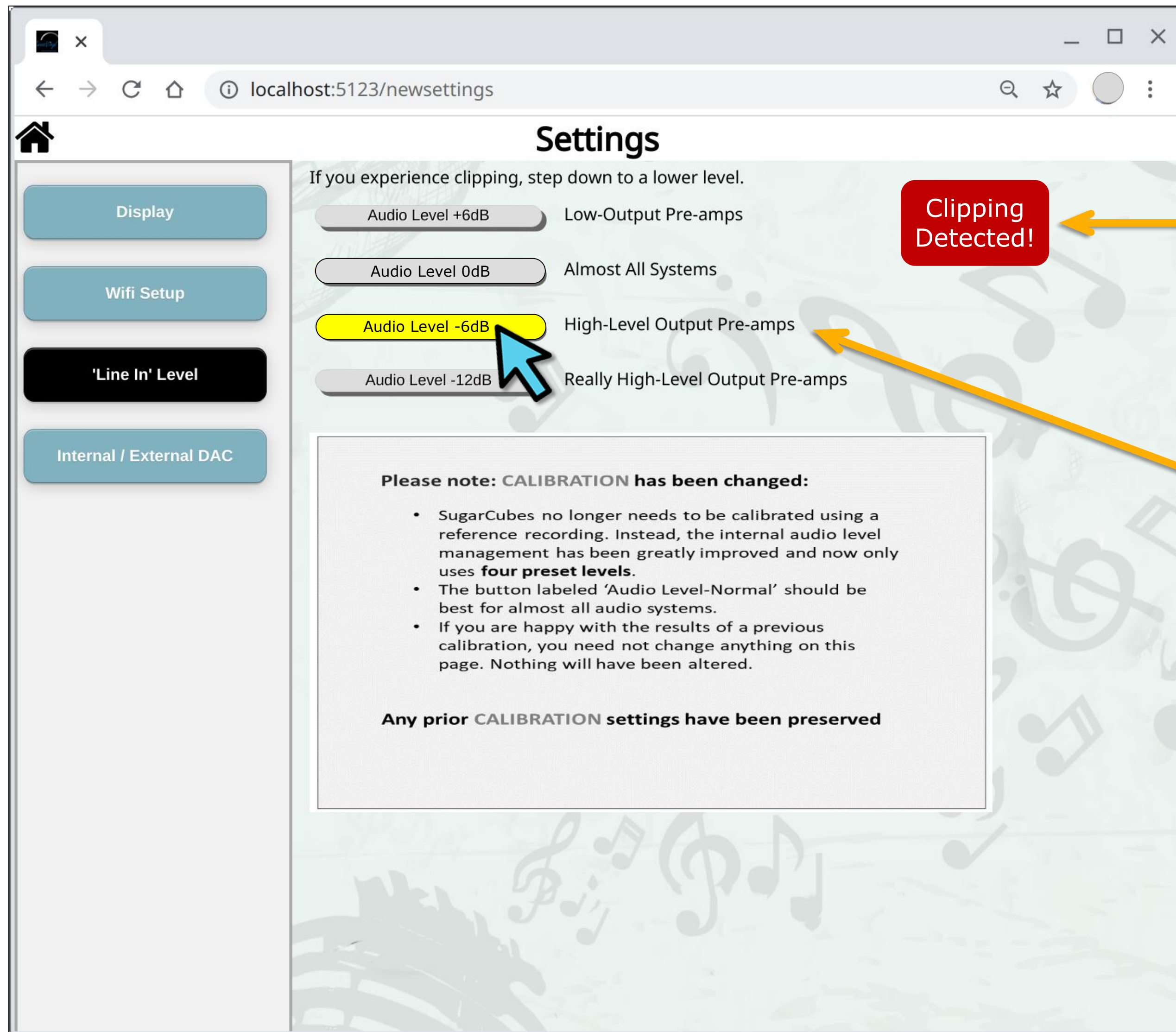
Select this mode **ONLY** if you know you have a very-low-output cartridge (and you have to turn up the volume higher than normal on your system.) If the 'Clipping' alert is seen, go back to 'Normal Output'.

Almost all systems should operate here (factory default).

If the 'Clipping' alert is seen on the app **AND** already in 'Normal Output' mode (above), select this.

If the 'Clipping' alert is seen on the app **AND** already in 'High Output' mode (above), select this.

42. Input Level Sensitivity



NOTE: if this appears when, for example, you just dropped the needle on an LP, just touch it and it will go away.

However: if you see this warning while an LP is playing, then move down a level.

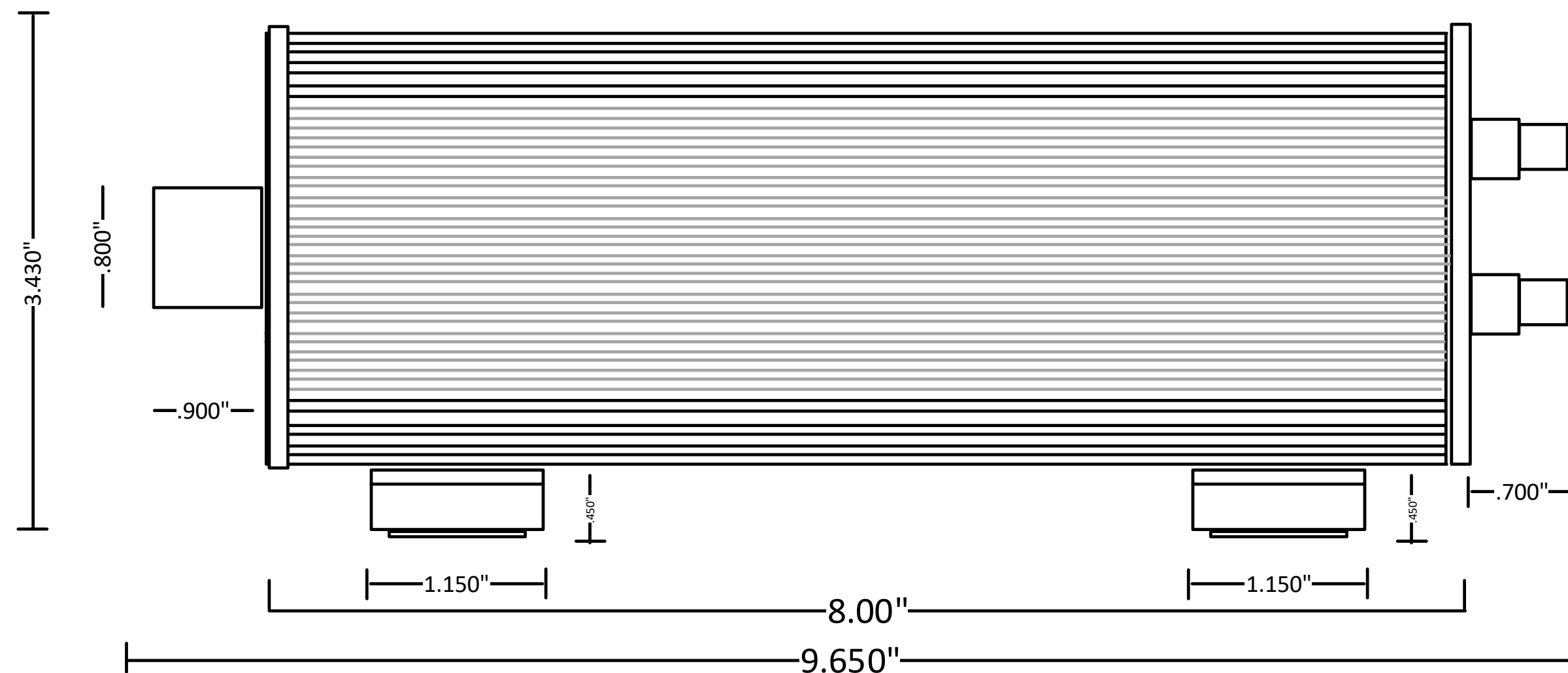
Technical Specs

Appendix A

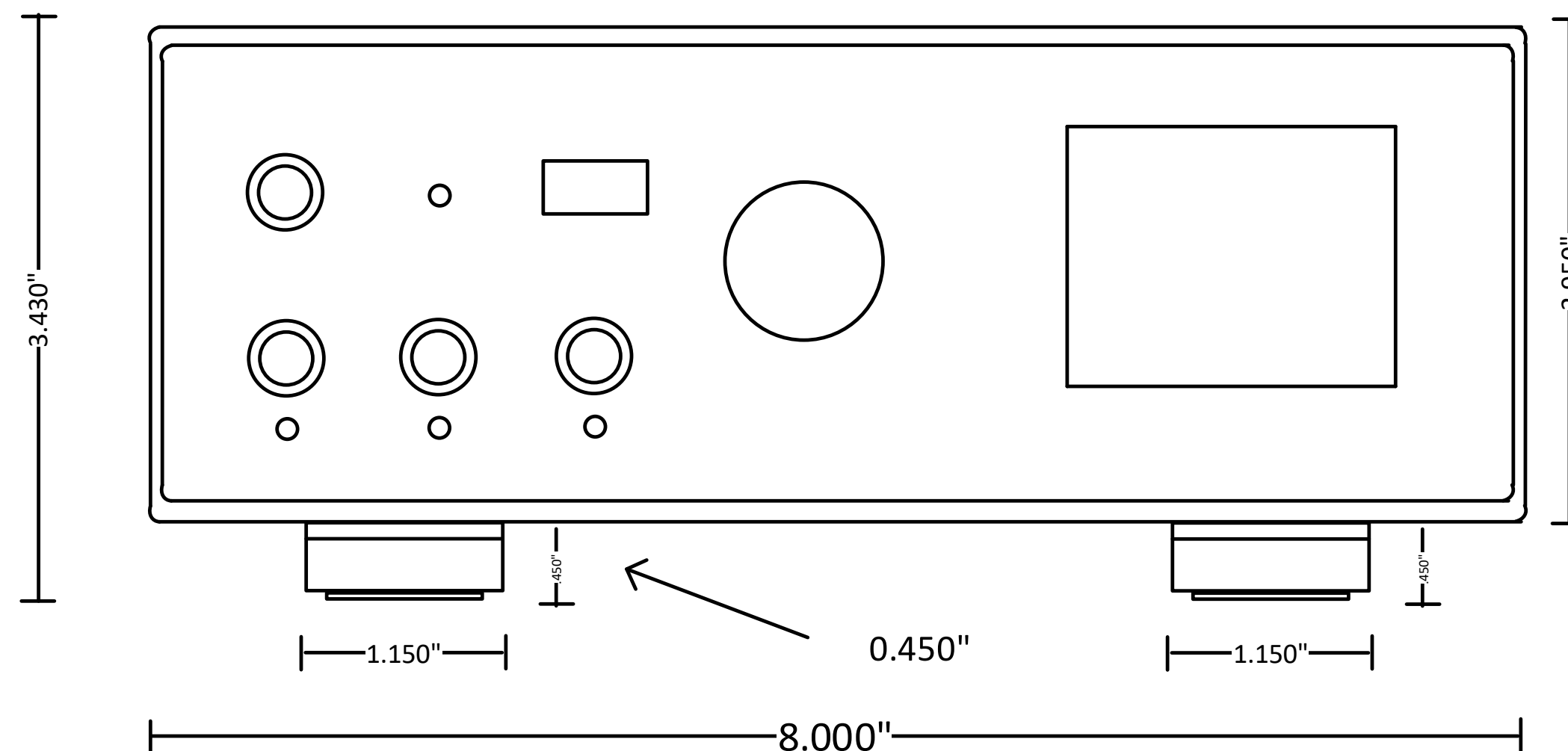


SugarCube SC2-Mini/Phono Mechanical:

SC-2 Mini - PS
Side View



Front View



Technical Specifications

Phono Input

- Gain: MM 40dB / MC 60dB
- Gain adjust: -6dB, 0dB, +6dB
- Input loading MM: 47k Ohms fixed and 100pF or 300pF selectable
- Input loading MC: 20 Ohms to 5k ohms, continuous adjustment
- Signal/noise ratio: >70dB 'A'-weighted Input
- Frequency response: 12Hz to 60kHz -3dB
- RIAA ± 1 dB, 20Hz to 20 kHz
- Rumble filter 3-pole, -18dB at 8Hz

General

- **Line In** maximum level = +20 dBu (approx. 7.7Vrms or 22Vp-p)
- **Line In** impedance = 100k Ohm
- **Line Out** level always equal to **Line In** level, regardless of any internal settings
- THD+N = 0.0030% (A-weighted)
- Frequency Response = 5Hz to 75kHz -3dB
- DC in = 5 volts, center-pin positive
- Average Power Dissipation = 8 watts

Support

Thank you for purchasing this SugarCube product.

Please contact customer support at: info@sweetvinyl.com

Trademarks

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Compliance

For the current EU Declaration of conformity, visit:
<http://support.sweetvinyl.com/regulatory>

For regulatory compliance information in North America, visit:
<http://support.sweetvinyl.com/regulatory>

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